Tanium™ Asset User Guide

Version 1.17.143

September 30, 2021

Table of contents

Asset overview

With Asset, you can get a complete and up-to-date view of your enterprise inventory. An asset is any endpoint that is managed by a Tanium Client, such as a computer, server, or virtual machine. The inventory is an aggregation of live asset data with the most recent data from offline assets.

You can build reports to show an overview of all assets, or you can drill down into specific endpoints or users.

Sources

Sources define where Asset gets data from. By default, Asset runs scheduled saved questions in Tanium to populate information about endpoints in the Asset database.

You can augment Asset inventory data with data from a SQL Server database. By importing data that is typically not available on an endpoint into the Asset data store, you can enable filtering and reporting on information such as department, cost center, building, and location.

Online and offline asset data aggregation

Asset aggregates data from both online and offline assets into one complete reporting data set. This data is populated from saved questions that are scheduled on a configurable basis. All endpoints that have answered Tanium Asset questions are visible in Asset reports. You can configure how long to keep data for endpoints that have not answered Tanium Asset questions.

Predefined reports

Asset comes with a set of predefined reports to help you prepare for audit and inventory activities.

- All Assets
- Physical Machine Summary
- Virtual Machine Summary
- Platform Summary
- All Users
- All Software
- All Microsoft Software
- All Adobe Software
- Age of Assets
- Age of Assets for Lost Devices
- Age of Assets for New Devices

- SIU Installed Products
- SIU Product Usage
- SIU Product Last Used

Summary report

Summary reports group data by attributes that you select and display a count of matching endpoints.

You can drill down on the endpoint count and see a filtered list of computer details, which includes the software and hardware information about the asset.

If the report contains computer name or user name, you can also drill down for further details, such as operating system and hardware information.

Custom reports

You can build custom reports that are based on any existing report. You can also create your own reports from existing Tanium sensor data or custom content. You can create custom reports to show assets by department, location, user group, or other attributes.

Views

You can build an alternative perspective of the Asset data with views. Views specify available attributes and can filter the included data. You can export data from a view to Tanium Connect.

Software inventory and usage

You can track and monitor software inventory and usage on endpoints. To enable usage reporting of a product in Asset, enable tracking on the product.

Data export

Tanium™ Connect

You can use any predefined reports, custom reports, or views as a connection source and send to destinations such as Email, File, HTTP, Socket Receiver, Splunk, and SQL Server.

Flexera

You can use the existing Tanium Client to populate information in Flexera FlexNet Manager Suite (FNMS).

ServiceNow CMDB

You can define the server, attribute mappings, and schedule for data to be exported to ServiceNow CMDB, ensuring that it has up-to-date information.

Integration with other Tanium products

Asset has built in integration with other Tanium products to provide additional features and reporting.

Tanium™ Trends

Asset features Trends boards that provide data visualization of Asset concepts. The **Asset** board displays the number of endpoints reporting to Asset, and how the data is being used in reports, views, and data exports. The following panels are in the **Asset** board:

- Total number of endpoints
- Number of endpoints seen today
- Age of assets
- ServiceNow insights
- Report operations
- Number of reports
- Number of views
- Average job duration

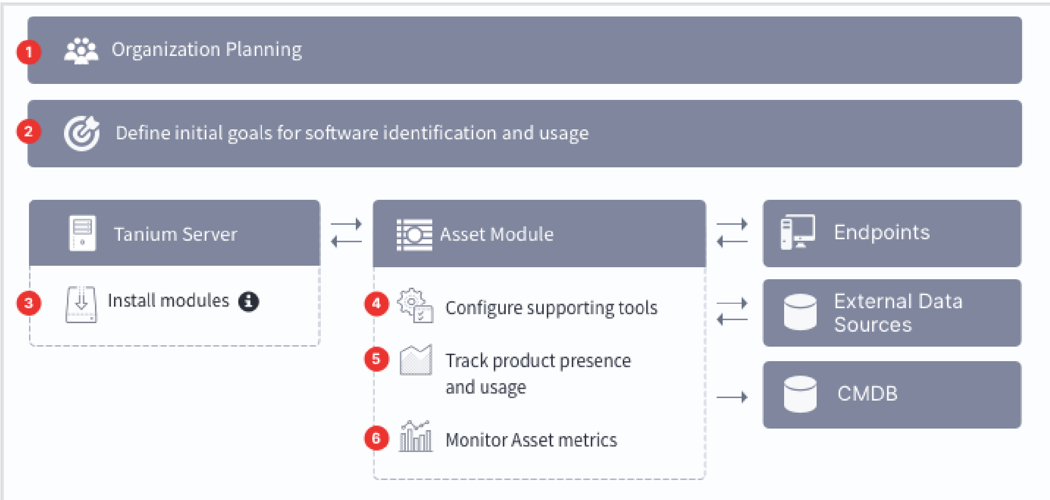
For more information about how to import the Trends boards that are provided by Asset, see Tanium Trends User Guide: Importing the initial gallery.

Tanium Connect

With Tanium Connect, you can send data to destinations, such as Email, File, HTTP, Socket Receiver, Splunk, and SQL Server.

Succeeding with Asset

Follow these best practices to achieve maximum value and success with Tanium Asset. These steps align with the key benchmark metrics: age of assets, frequency of imports and exports, key vendors with tracked products, and unused tracked products.



Step 1: Gain organizational effectiveness

☐ Develop a dedicated Change management on page 15 process for the implementation and maintenance of Tanium Asset.

☐ Define distinct roles and responsibilities in a RACI chart on page 15 for change management.

☐ Validate cross-functional Organizational alignment on page 17.

☐ Track Operational metrics on page 17.

Step 2: Define initial goals for software identification and usage

☐ Analyze existing policies and procedures for software usage and license reclamation.

☐ Choose high-cost or high-impact products to inventory and analyze.

Step 3: Install Tanium modules

☐ Install Tanium Client Management, which provides Tanium Endpoint Configuration. See Tanium Client Management User Guide: Installing Client Management.

☐ Install Tanium Trends. See Tanium Trends User Guide: Installing Trends.

☐ Install Tanium Connect. See Tanium Connect User Guide: Installing Connect.

☐ Install Tanium Asset. See Installing Asset on page 29.

When you import Asset with automatic configuration, the following default settings are configured:

Setting	Default value
Action group	• Restricted targeting disabled (default): `All Computers` computer group • Restricted targeting enabled: `No Computers` computer group
Service account	The service account is set to the account that you used to import the module. Configuring a unique service account for each Tanium solution is an extra security measure to consider in consultation with the security team of your organization. See Configure service account on page 32.
Import schedule	The import schedule is set to start collecting data and generating reports.

☐ To locate file-based evidence, install Tanium Index and verify that endpoint file systems are being indexed. Deploy the **Asset Start File Evidence Scan** and **Asset Collect MS Exchange Info** packages to your endpoints. For more information, see Tanium Incident Response User Guide: Install Index and Tanium Incident Response User Guide: Deploy Index tools.

NOTE

Do not install Index if you are using Tanium Threat Response. Threat Response automatically installs Index.

Step 4: Configure supporting tools

☐ To support primary user tracking, install the Core AD Query Content solution. See Installing Asset on page 29.

☐ Enable scheduled actions for software inventory and usage and Flexera file evidence content. See Installing Asset on page 29.

Step 5: Track product presence and usage

Begin tracking product presence and usage to meet stated goals.

☐ Review products discovered by Asset. For more information, see View software inventory and usage on page 62

☐ Track products in accordance with organizational priorities. For more information, see Track a product to enable reporting on page 61

☐ Plan software usage analysis with stakeholders.

Step 6: Monitor Asset metrics

☐ From the Asset Overview page, review the **Age of Assets** panel.

☐ To understand the frequency of imports and exports, review **Inventory Management > Schedules**.

☐ In **Software Inventory & Usage > All Vendors**, review the key vendors with tracked products. For more information, see Monitor and troubleshoot key vendors with tracked products on page 84.

☐ In **Software Inventory & Usage > All Products**, identify the unused tracked products. For more information, see Monitor and troubleshoot unused tracked software on page 84.

Gaining organizational effectiveness

The four key organizational governance steps to maximizing the value that is delivered by Asset are as follows:

- Develop a dedicated change management process. See Change management on page 15.
- Define distinct roles and responsibilities. See RACI chart on page 15.
- Track operational maturity. See Operational metrics on page 17.
- Validate cross-functional alignment. See Organizational alignment on page 17.

Change management

Develop a tailored, dedicated change management process for asset management, taking into account the new capabilities provided by Tanium.

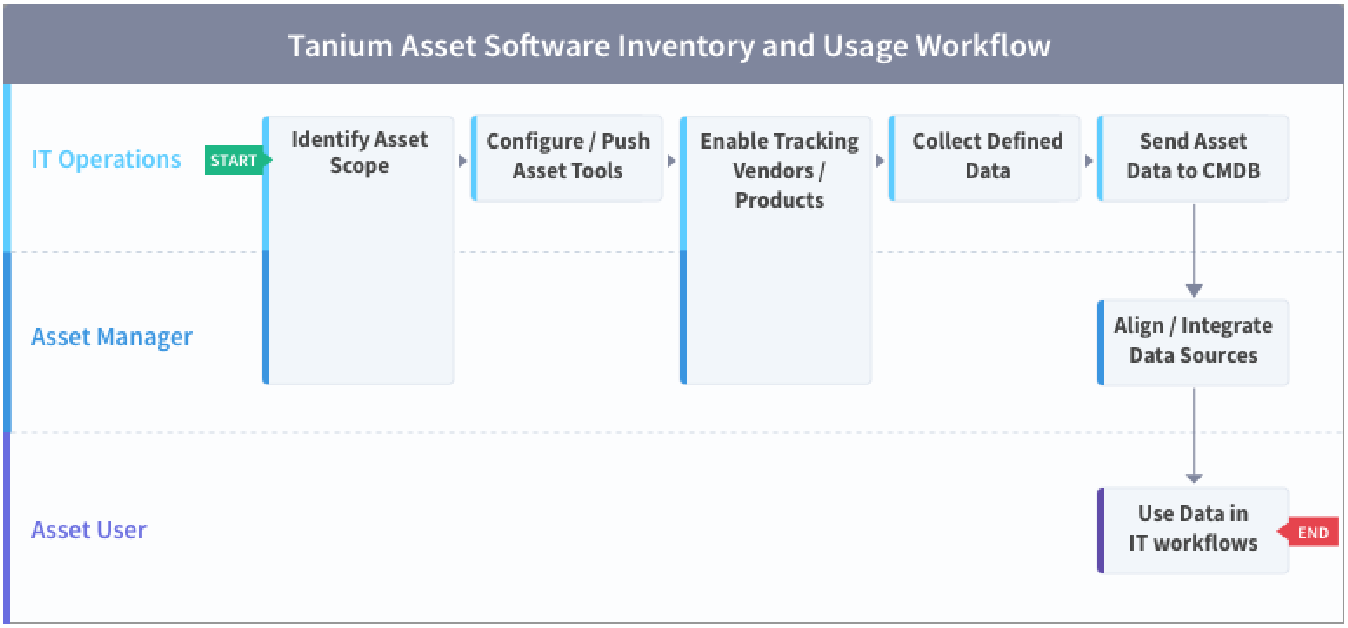
- Update SLAs and align activities to key resources for Tanium Asset activities across IT Security, IT Operations, and IT Risk / Compliance.
- Designate change or maintenance windows for various asset discovery scenarios (example: setting up Active Directory Query and external data sources, integration with CMDBs such as ServiceNow or Flexera).
- Identify internal and external dependencies to your asset discovery process (example: to achieve effective integrations with the CMDB or Active Directory Query).
- Create a Tanium Steering Group (TSG) for asset management activities to expedite reviews and approvals of processes that align with SLAs.
- Determine frequency of reviews for changes and new information.
- Consider conducting at least a quarterly review of change management process.

RACI chart

A RACI chart identifies the team or resource who is **R**esponsible, **A**ccountable, **C**onsulted, and **I**nformed, and serves as a guideline to describe the key activities across the security, risk/compliance, and operations teams. Every organization has specific business processes and IT organization demands. The following table represents Tanium's point of view for how organizations should align functional resources against asset management. Use the following table as a baseline example.

Task	IT Operations	Asset manager	Asset user	Rationale
Identify Asset scope	R	C	C	The Tanium platform owner must identify computers that are in scope for Asset. This scope is likely all computers, but a conversation should occur between the CMDB owners and the Tanium platform owners to fully understand scope across both tool sets.
Configure supporting tools	R	-	-	The Tanium platform owner rolls out the required Asset Tools (including Indexing capability) to the determined set of computers. The Tanium platform owner also must continually monitor tool deployment and upgrades.
Create Asset reports	R	R / C	R / C	The Tanium platform owner and business process owner end user must define the report requirements, then build those reports in Asset for viewing by business owners. The platform owner can configure role-based access control (RBAC) to allow users to view reports, or allow business owners to build read-only reports.
Asset RBAC management	R	C	I	The Tanium platform owner must plan and implement appropriate Role Based Access Controls to allow other business units to access and create reports and views.
Enable tracking of products	C	R	C	If Software Inventory & Usage is configured, the platform owner can enable the business owners to align to the identified vendors and products in Asset or the CMDB.
Collect defined data	I	I	I	Tanium Asset collects asset data based on defined requirements and configurations.

Software inventory and usage workflow



Organizational alignment

Successful organizations use Tanium across functional silos as a common platform for high-fidelity endpoint data and unified endpoint management. Tanium provides a common data schema that enables security, operations, and risk/compliance teams to assure that they are acting on a common set of facts that are delivered by a unified platform.

In the absence of cross-functional alignment, functional silos often spend time and effort in litigating data quality instead of making decisions to manage software and hardware assets.

Operational metrics

Asset maturity

Managing an asset management program successfully includes operationalization of the technology and measuring success through key benchmarking metrics. The key processes to measure and guide operational maturity of your Tanium Asset program are as follows:

Process	Description
Usage	How and when Tanium Asset is used in your organization
Automation	How automated Tanium Asset is, across endpoints
Reporting	How data from Asset is consumed by people and systems within the organization

Benchmark metrics

In addition to the key asset processes, the key benchmark metrics that align to the operational maturity of the Tanium Asset program to achieve maximum value and success are as follows:

Executive Metrics	Key Vendors with Tracked Products	Unused Tracked Products
Description	The key vendors with tracked products	The tracked products that are not being used
Instrumentation	**Software Inventory & Usage > All Vendors**	**Software Inventory & Usage > All Products**
Why this metric matters	Key vendors with a cost associated with them that do not have any tracked products are a detriment to the value provided by Asset, especially when it comes to software utilization. Cost-significant products should be tracked.	The metric demonstrates the usage and effectiveness of software the company has determined is worth tracking on a per user level. Unused software licenses can be reclaimed.

Use the following table to determine the maturity level for Tanium Asset in your organization.

		Level 1 (Needs improvement)	Level 2 (Below average)	Level 3 (Average)	Level 4 (Above average)	Level 5 (Optimized)
Process	Usage	• Asset installed • Data retention policies and access established	• Primary user(s) identified on endpoints • Endpoint use case gap analysis complete	Custom sensors created for use, and new attributes added from custom sensors	Destinations (ServiceNow or Flexera) and external sources are configured, providing enrichment and context to Asset endpoint data	• Destinations (ServiceNow or Flexera) and external sources are configured, providing enrichment and context to Asset endpoint data • All endpoint data flows through Asset, either through imports or exports

		Level 1 (Needs improvement)	Level 2 (Below average)	Level 3 (Average)	Level 4 (Above average)	Level 5 (Optimized)
	Automation	Import schedule configured, assets are visible	• Integration with Active Directory. Active Directory Query content imported • Data retention rules enabled and aligned with IT policies • Exporting data from report grid view	• Asset views created to export data • User group permission enabled and aligned with IT policies • Evaluation of ServiceNow or Flexera CMDB for enrichment using Tanium Asset	• Sending additional attributes to CMDB (ServiceNow or Flexera) • Sending hardware, application, and primary user details to Splunk / SQL Server or Elastic	• Sending additional attributes to CMDB (ServiceNow or Flexera) • Sending hardware, application, and primary user details to Splunk / SQL Server or Elastic • Achieve custom Asset integration with third-party REST endpoint with Tanium Connect
	Reporting	Manual; Asset workbench and dashboard for Operators only	Manual; Asset reports using integration details	Automated; exporting scheduled reports to file / email / SQL	Automated; CMDBs receiving data from Tanium Asset for reporting	Automated; CMDBs receiving data from Tanium Asset for reporting, including any third-party products with the Tanium Asset API
Metrics	Key Vendors with Tracked Products	70%	80%	90%	95%	98%
	Unused Tracked Products	50%	40%	30%	20%	5%

Asset requirements

Review the requirements before you use Asset.

Tanium dependencies

Component	Requirement
Tanium™ Core Platform	• 7.3.314.4250 or later • TanOS 1.3.4 or later
Tanium™ Content	(Optional) Asset includes all of the content it needs for base functionality. You can import additional content or sensors into Asset after installation.
Tanium™ Client	Any supported version of Tanium Client. For the Tanium Client versions supported for each OS, see Tanium Client Management User Guide: Client version and host system requirements. If you use a client version that is not listed, certain product features might not be available, or stability issues can occur that can only be resolved by upgrading to one of the listed client versions.
Tanium solutions	If you selected **Tanium Recommended Installation** when you installed Asset, the Tanium Server automatically installed all your licensed solutions at the same time. Otherwise, you must manually install the solutions that Asset requires to function, as described under Tanium Console User Guide: Import, re-import, or update specific solutions. Asset requires the following Tanium solutions: • Tanium Endpoint Configuration 1.2 or later (installed as part of Tanium Client Management 1.5 or later) • Tanium Trends 3.6 or later (create charts on Asset Overview page) The following solutions are optional: • Tanium Connect 4.3 or later (create connections with Asset reports as a data source) • Tanium Index (create reports with file evidence data, for example, the Flexera File Evidence report)

Tanium™ Module Server

Asset runs as a service on the Tanium Module Server.

Disk space

Asset requires disk storage capacity that is necessary to support the number of endpoints in the environment. For planning purposes, use 100 MB per 1000 endpoints:

- 5,000 endpoints: 500 MB
- 50,000 endpoints: 5 GB
- 100,000 endpoints: 10 GB
- 250,000 endpoints: 25 GB
- 500,000 endpoints: consult your Technical Account Manager

Usage might vary significantly based on the following variables:

- Number of endpoints
- Number of applications
- Number of users, if file evidence data is enabled
- Attributes that you add on the **Inventory Management > Attributes** page.

These suggested sizes are considered a good estimate for most environments.

Endpoints

Supported internet protocols

Asset communicates over IPv4 and IPv6 networks. For more information, see Tanium Client Management User Guide: Network connectivity, ports, and firewalls.

Supported operating systems

For Tanium Client operating system support, see Tanium Client Management User Guide: Client version and host system requirements.

Operating System	Version
Windows	• Windows 7 SP1 or later • Windows 2008 R2 SP1 or later
macOS	Same as Tanium Client support
Linux	Same as Tanium Client support NOTE: Software Inventory & Usage is not available on Linux operating systems.

Operating System	Version
Solaris	Same as Tanium Client support NOTE: Software Inventory & Usage is not available on Solaris operating systems.
AIX	7.1.4 or later The IBM XL C++ runtime libraries file set (`xlC.rte`), version 16.1.0.0 or later, and the IBM LLVM runtime libraries file set (`libc++.rte`) must be installed. For installation instructions, see Tanium Client Management User Guide: Deploy the Tanium Client to AIX endpoints using a package file. NOTE: Software Inventory & Usage is not available on AIX operating systems.

Third-party software

The following third-party software is optional:

- For the ServiceNow CMDB connector to export data from Asset, the Jakarta release or later is required.
- For Flexera integration to export data from Asset, you must have an SQL database that can be configured to receive data from Asset. For more information, Contact Tanium Support on page 85.
- To use the **Asset Collect MS Exchange Info** package to collect Microsoft Exchange data, the Microsoft Exchange Server Computer objects need to be a member of the **View-Only Organization Management** group for **Microsoft Exchange Security Groups**.

Host and network security requirements

Specific ports and processes are needed to run Asset.

Ports

The following ports are required for Asset communication.

Source	Destination	Port	Protocol	Purpose
Module Server	ServiceNow	443	TCP	Access to your ServiceNow instance
Module Server	Module Server (loopback)	17459 17461	TCP	Internal purposes; not externally accessible

BEST PRACTICE Configure firewall policies to open ports for Tanium traffic with TCP-based rules instead of application identity-based rules. For example, on a Palo Alto Networks firewall, configure the rules with service objects or service groups instead of application objects or application groups.

Security exclusions

A security administrator must create exclusions to allow Tanium processes to run without interference if security software is in use in the environment to monitor and block unknown host system processes.

Asset security exclusions

Target Device	Notes	Exclusion Type	Exclusion
Module Server		Process	`<Module Server>\services\asset-service\node.exe`
		Process	`<Module Server>\services\asset-service\node_modules@tanium\postgresql\lib\win32\bin\postgres.exe`
		Process	`<Module Server>\services\asset-service\node_modules@tanium\postgresql\lib\win32\bin\pg_ctl.exe`
		Process	`<Module Server>\services\endpoint-configuration-service\TaniumEndpointConfigService.exe`
Windows endpoints	For integration with Flexera	Process	`<Tanium Client>\Tools\EPI\TaniumEndpointIndex.exe`
		Process	`<Tanium Client>\Tools\Asset\TaniumFileEvidence.exe`
		Process	`<Tanium Client>\extensions\TaniumSoftwareManager.dll`
		Process	`<Tanium Client>\extensions\TaniumSoftwareManager.dll.sig`
macOS endpoints	For integration with Flexera	Process	`<Tanium Client>/Tools/EPI/TaniumEndpointIndex`
		Process	`<Tanium Client>/Tools/Asset/TaniumFileEvidence`
		Process	`<Tanium Client>/extensions/libTaniumSoftwareManager.dylib`
		Process	`<Tanium Client>/extensions/libTaniumSoftwareManager.dylib.sig`
Linux endpoints	For integration with Flexera	Process	`<Tanium Client>/Tools/EPI/TaniumEndpointIndex`
		Process	`<Tanium Client>/Tools/Asset/TaniumFileEvidence`
		Process	`<Tanium Client>/extensions/libTaniumSoftwareManager.so`
		Process	`<Tanium Client>/extensions/libTaniumSoftwareManager.so.sig`

Internet URLs

If security software is deployed in the environment to monitor and block unknown URLs, a security administrator might need to allow the following URLs on the Tanium Module Server for the Asset service.

- ServiceNow instance (yourcompany.service-now.com)

User role requirements

The following tables list the role permissions required to use Asset. To review a summary of the predefined roles, see Set up Asset users on page 33.

For more information about role permissions and associated content sets, see Tanium Console User Guide: Managing RBAC.

Asset user role permissions

Permission	Asset Administrator[1]	Asset Operator[1]	Asset User[1]	Asset Report Reader[1]	Asset Service Account[1,5]	Asset Endpoint Configuration Approver[2]
Asset View Asset workbench	✔ SHOW	✔ SHOW	✔ SHOW	✔ SHOW	✖	✔ SHOW
Asset Configuration Item Configure all aspects of Asset (service settings, schedules, attributes, destinations)	✔ WRITE	✔ WRITE	✖	✖	✖	✖
Asset Endpoint Configuration Approve Asset configuration changes in the Endpoint Configuration service	✖	✖	✖	✖	✖	✔ APPROVE
Asset Plugin Configure Asset communication with the Tanium Server and Tanium Module Server	✖	✖	✖	✖	✔ CALLBACK	✖

Asset user role permissions (continued)

Permission	Asset Administrator[1]	Asset Operator[1]	Asset User[1]	Asset Report Reader[1]	Asset Service Account[1,5]	Asset Endpoint Configuration Approver[2]
Asset Report View, create, edit, and delete reports and views	✓ READ WRITE	✓ READ WRITE	✓ READ WRITE[3]	✓ READ	✗	✗
Asset Service Configure all aspects of Asset services	✓ CONFIGURE	✗	✗	✗	✗	✗
Asset Trends Integration Service Account Provide access for module service accounts to read and write data, and to define sources and boards	✗	✓ EXECUTE[4]	✗	✗	✓ EXECUTE[4]	✗

[1] This role provides module permissions for Tanium Trends. You can view which Trends permissions are granted to this role in the Tanium Console. For more information, see Tanium Trends User Guide: User role requirements.

[2] This role provides module permissions for Tanium Endpoint Configuration. You can view which Endpoint Configuration permissions are granted to this role in the Tanium Console. For more information, see the Tanium Endpoint Configuration User Guide: User role requirements.

[3] For owned reports and views only.

[4] Grants access to content in the Reserved content set.

[5]If you installed Tanium Client Management, Endpoint Configuration is installed, and by default, configuration changes initiated by the module service account (such as tool deployment) require approval. You can bypass approval for module-generated configuration changes by applying the **Endpoint Configuration Bypass Approval** permission to this role and adding the relevant content sets. For more information, see Tanium Endpoint Configuration User Guide: User role requirements.

Provided Asset administration and platform content permissions

Permission	Permission Type	Asset Administrator[1]	Asset Operator[1]	Asset User[1]	Asset Report Reader[1]	Asset Service Account[1]	Asset Endpoint Configuration Approver[1]
Action Group	Administration	✓ READ WRITE	✓ READ WRITE	✓ READ	✓ READ	✓ READ WRITE	✗
Action	Platform Content	✓ READ WRITE	✓ READ WRITE	✗	✗	✓ WRITE	✗
Action for Saved Question	Platform Content	✗	✗	✗	✗	✓ WRITE	✗
Filter Group	Platform Content	✓ READ	✓ READ	✗	✗	✗	✗
Own Action	Platform Content	✓ READ	✓ READ	✗	✗	✓ READ	✗
Package	Platform Content	✓ READ	✓ READ	✗	✗	✓ READ	✗
Plugin	Platform Content	✓ READ EXECUTE	✓ READ EXECUTE	✓ READ EXECUTE	✓ READ EXECUTE	✓ READ EXECUTE	✓ READ EXECUTE
Saved Question	Platform Content	✗	✗	✗	✗	✓ READ WRITE	✗
Sensor	Platform Content	✓ READ	✓ READ	✓ READ	✓ READ	✓ READ	✗

You can view which content sets are granted to any role in the Tanium Console.

[1] This role provides content set permissions for Tanium Trends. You can view which Trends permissions are granted to this role in the Tanium Console. For more information, see Tanium Trends User Guide: User role requirements.

Optional roles for Asset

Role	Enables
Connect Administrator (prior to Connect 4.8 only)	Create, edit, or delete a Flexera destination
Connect User (Connect 4.8 and later)	Create, edit, or delete a Flexera destination
Tanium Administrator	Create scheduled actions for the file evidence content for Flexera destinations

Installing Asset

Use the **Solutions** page to install Asset and choose either automatic or manual configuration:

- **Automatic configuration with default settings** (Tanium 7.4.2 and later only): Asset is installed with any required dependencies and other selected products. After installation, the Tanium Server automatically configures the recommended default settings. This option is the best practice for most deployments. For more information about the automatic configuration for Asset, see Import Asset with default settings on page 29.
- **Manual configuration with custom settings**: After installing Asset, you must manually configure required settings. Select this option only if Asset requires settings that differ from the recommended default settings. For more information, see Import Asset with custom settings on page 30.

Before you begin

- Read the release notes.
- Review the Asset requirements on page 21.

Import Asset with default settings

When you import Asset with automatic configuration, the following default settings are configured:

Setting	Default value
Action group	• Restricted targeting disabled (default): `All Computers` computer group • Restricted targeting enabled: `No Computers` computer group
Service account	The service account is set to the account that you used to import the module. Configuring a unique service account for each Tanium solution is an extra security measure to consider in consultation with the security team of your organization. See Configure service account on page 32.
Import schedule	The import schedule is set to start collecting data and generating reports.

To import Asset and configure default settings, be sure to select the **Apply All Tanium recommended configurations** check box while performing the steps in Tanium Console User Guide: Import all modules and services. After the import, verify that the correct version is installed: see Verify Asset version on page 30.

(Tanium Core Platform 7.4.5 or later only) You can set the Asset action group to target the **No Computers** filter group by enabling restricted targeting before importing Asset. This option enables you to control tools deployment through scheduled actions that are created during the import and that target the Tanium Asset action group. For example, you might want to test tools on a subset of endpoints before deploying the tools to all endpoints. In this case, you can manually deploy the tools to an action group that you configured to target only the subset. To configure an action group, see Tanium Console User Guide: Managing action groups. To enable or disable restricted targeting, see Tanium Console User Guide: Dependencies, default settings, and tools deployment.

Import Asset with custom settings

To import Asset without automatically configuring default settings, be sure to clear the **Apply All Tanium recommended configurations** check box while performing the steps in Tanium Console User Guide: Import, re-import, or update specific solutions. After the import, verify that the correct version is installed: see Verify Asset version on page 30.

To configure the service account, see Configure service account on page 32.

Manage dependencies for Tanium solutions

When you start the Asset workbench for the first time, the Tanium Console ensures that all of the required dependencies for Asset are installed at the required version. You must install all required Tanium dependencies before the Asset workbench can load. A banner appears if one or more Tanium dependencies are not installed in the environment. The Tanium Console lists the required Tanium dependencies and the required versions.

1. Install the modules and shared services that the Tanium Console lists as dependencies, as described under Tanium Console User Guide: Import, re-import, or update specific solutions.
2. From the Main menu, go to **Modules > Asset** to open the Asset **Overview** page.

Upgrade Asset

For the steps to upgrade Asset, see Tanium Console User Guide: Manage Tanium modules. After the upgrade, verify that the correct version is installed: see Verify Asset version on page 30.

Verify Asset version

After you import or upgrade Asset, verify that the correct version is installed:

1. Refresh your browser.
2. From the Main menu, go to **Modules > Asset** to open the Asset **Overview** page.
3. To display version information, click Info .

Configuring Asset

If you did not install Asset with the **Apply All Tanium recommended configurations**, you must enable and configure certain features.

Enable collection of Active Directory (AD) information

(Optional) To gather user data from Windows endpoints, including Full Name, Email, Phone, Department, and Location, install the Active Directory Query solution and create a scheduled action for the **Collect Active Directory Info** package.

For Mac and Linux endpoints, you do not need to deploy any actions to get this information.

1. Install the Active Directory Query solution.
   a. From the Main menu, go to **Administration > Solutions**.
   b. In the **Tanium Content** section, select the **Core AD Query Content** row and click **Import Solution**.
   c. Review the list of packages and sensors and click **Proceed with Import**.
2. Run the **Collect Active Directory Info** package on your endpoints with a scheduled action. For more information, see [Tanium Core Platform User Guide: Managing Scheduled Actions](#).
   a. Use the **Is Windows** sensor to target Windows endpoints.
   b. Deploy the **Collect Active Directory Info** package to your endpoints. Configure a saved action as a scheduled action. Set **Distribute Over** to `1 hour`, and set **Reissue Every** to `3 hours`.

The action gets recent sign ins and the primary user of each system. A primary user has the most interactive sign-ins the past 30 days. You can access this information with the **Primary User Details** sensor or by adding the **User Name** column to a report.

Prepare endpoints

Manage solution configurations with Tanium Endpoint Configuration

Tanium Endpoint Configuration delivers configuration information and required tools for Tanium Solutions to endpoints. Endpoint Configuration consolidates the configuration actions that traditionally accompany additional Tanium functionality and eliminates the potential for timing errors that occur between when a solution configuration is made and the time that configuration reaches an endpoint. Managing configuration in this way greatly reduces the time to install, configure, and use Tanium functionality, and improves the flexibility to target specific configurations to groups of endpoints.

> NOTE
> Endpoint Configuration is installed as a part of Tanium Client Management. For more information, see the Tanium Client Management User Guide: Installing Client Management.

Additionally you can use Endpoint Configuration to manage configuration approval. For example, configuration changes are not deployed to endpoints until a user with approval permission approves the configuration changes in Endpoint Configuration. For more information about the roles and permissions that are required to approve configuration changes for Asset, see User role requirements on page 25.

To use Endpoint Configuration to manage approvals, you must enable configuration approvals.

1. From the Main menu, go to **Administration > Shared Services > Endpoint Configuration** to open the Endpoint Configuration **Overview** page.
2. Click Settings and click the **Global** tab.
3. Select **Enable configuration approvals**, and click **Save**.

For more information about Endpoint Configuration, see Tanium Endpoint Configuration User Guide.

Configure Asset

Configure service account

The service account is a user that runs several background processes for Asset. This user requires the following roles and access:

- **Tanium Administrator** or **Asset Service Account** role

  If you use the Asset Service Account role, be sure that the Asset service account user has sufficient computer group rights for visibility of Asset inventory. For example, for complete visibility, assign the service account user **Unrestricted Management Rights** for computer group assignments. For more information, see Tanium Console User Guide: Manage computer group assignments for a user.
- If you installed Tanium Client Management, Endpoint Configuration is installed, and by default, configuration changes initiated by the module service account (such as tool deployment) require approval. You can bypass approval for module-generated configuration changes by applying the **Endpoint Configuration Bypass Approval** permission to this role and adding the relevant content sets. For more information, see Tanium Endpoint Configuration User Guide: User role requirements.

For more information about Asset permissions, see User role requirements on page 25.

> NOTE
>
> If you imported Asset with default settings, the service account is set to the account that you used to perform the import. Configuring a unique service account for each Tanium solution is an extra security measure to consider in consultation with the security team of your organization.

1. From the Main menu, go to **Modules > Asset** to open the Asset **Overview** page.
2. Click Settings and open the **Service Account** tab.
3. Update the service account settings and click **Save**.

Configure Asset action group for software inventory and usage and Flexera file evidence content

(Optional) Asset includes sensors for software inventory and usage and Flexera. To use these features, deploy this content to the endpoints. The Tanium Asset action group and associated actions to deploy tools and distribute configurations are created by Tanium Endpoint Configuration. By default, the computer group target for the Tanium Asset action group includes all computers. You can change the target computer group, if necessary.

For more information about Flexera integration, see Flexera FlexNet Manager Suite: 2019 R1 or earlier on page 68.

1. From the Main menu, go to **Administration > Actions > Action Groups**.
2. In the list of action groups, click **Tanium Asset**.
3. Click **Edit**, update the action group to include a computer group that has the computers you want to include in your Asset data, and click **Save**.

Set up Asset users

You can use the following set of predefined user roles to set up Asset users.

To review specific permissions for each role, see User role requirements on page 25.

For more information about assigning user roles, see Tanium Core Platform User Guide: Manage role assignments for a user.

Asset Administrator

Assign the **Asset Administrator** role to users who manage all configuration and deployment of Asset functionality to endpoints.
This role can perform the following tasks:

- Configure all Asset service settings
- Manage the Tanium source
- View, create, edit, and delete all reports and views

Asset Operator

Assign the **Asset Operator** role to users who manage most configuration and deployment of Asset functionality to endpoints.
This role can perform the following tasks:

- Configure some Asset settings, including data retention, report timeout, and permissions
- View, create, edit, and delete all reports and views

Asset User

Assign the **Asset User** role to users who manage reports and views.
This role can view, create, edit, and delete owned reports and views.

Asset Report Reader

Assign the **Asset Report Reader** role to users who view reports and views.

Asset Service Account

Assign the **Asset Service Account** role to to an account that configures system settings for Asset.
This role can perform several background processes for Asset. For more information, see Installing Asset on page 29.

Asset Endpoint Configuration Approver

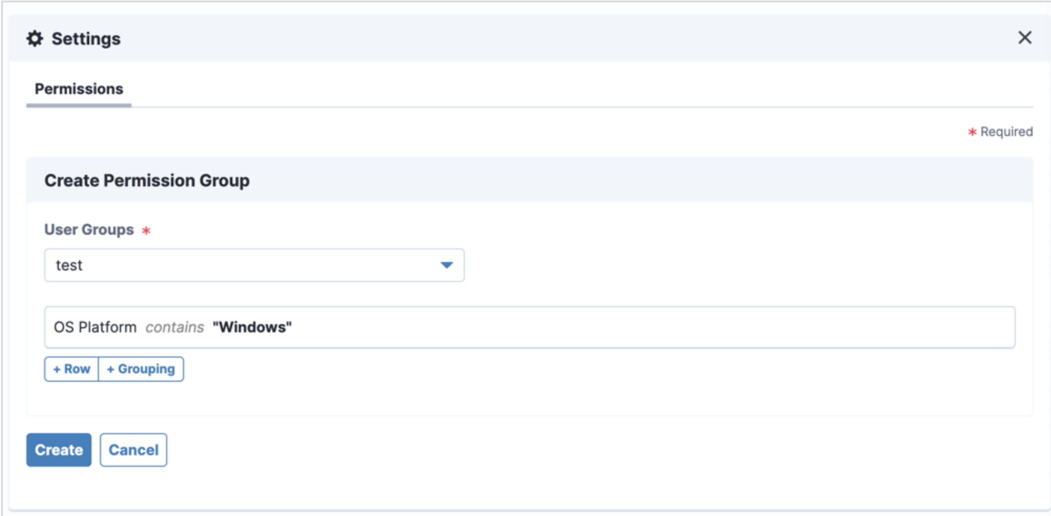
Assign the **Asset Endpoint Configuration Approver** role to a user who approves or rejects Asset configuration items in Tanium Endpoint Configuration.
This role can approve, reject, or dismiss changes that target endpoints where Asset is installed.

Set user permissions on computers

In addition to the Asset user roles that control access to Asset reports and settings as a whole, you can define more detailed permissions on computers that are based on individual attribute values. For example, you can create permissions that assign a user group permission to access information about Windows or macOS platform assets only. If a user belongs to multiple user groups, the permissions for all the user groups are combined with an OR operator.

Before you begin, you must have a user group to which you want to assign the Asset permissions. See Tanium Core Platform User Guide: Managing user groups. For the users in this user group to access Asset, they also must have an Asset user role assigned. See User role requirements on page 25.

1. From the Asset **Overview** page, click Settings ⚙. Click the **Permissions** tab.
2. Click **Create Permissions**.
3. Select the user group from the list that you want to assign.
4. Add a condition. This list of attributes is from the **ci_item** table of the Asset database.
   For example, to assign the user group permission to view Windows assets only, set to `OS Platform contains Windows`.



5. Click **Apply** to create each condition.

6. Click **Create**.

What to do next

See Succeeding with Asset on page 12 for more information about using Asset.

Building reports

You can build reports based on the existing built-in reports, or you can build your own reports from any sensor information that Tanium provides.

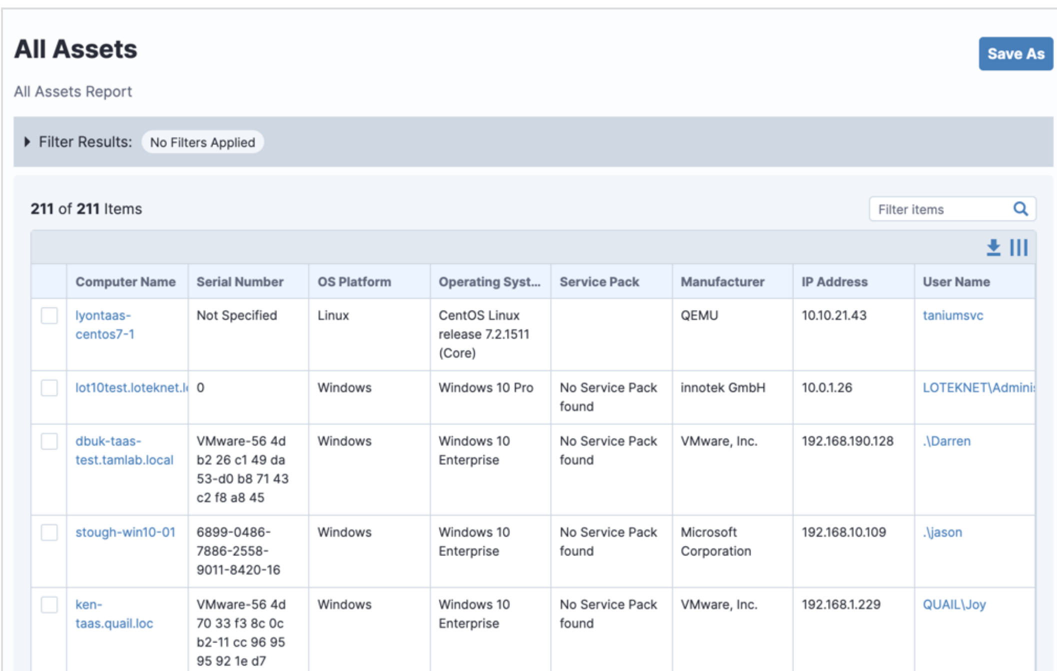
Work with existing reports

View reports

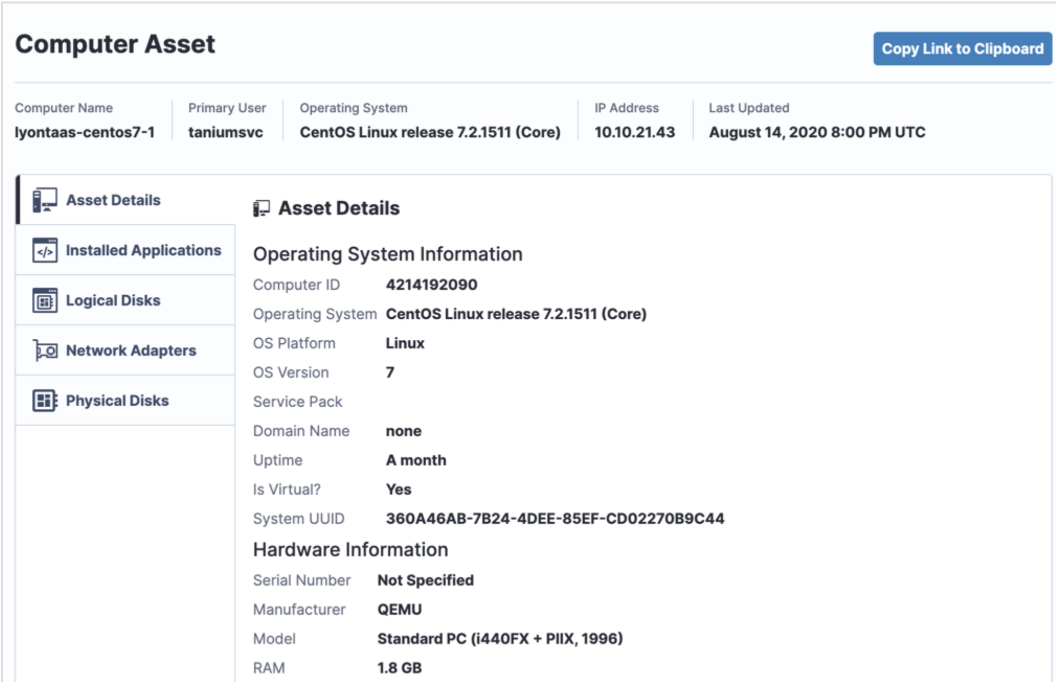
You view reports and assets within reports based on your permissions. For more information about how users receive permissions, see [Tanium Console User Guide: RBAC overview](#).

1. From the Asset menu, click **Reports**. All built-in and custom reports are displayed on this page. To view a report, click the report name.

2. If you are viewing a summary report, you can drill down into the data in a report by clicking links in the report data grid. For example, in the **All Assets** report, you can click the Computer Name to view information about that asset. You can also click the User Name to view information about assets that are owned by the user.



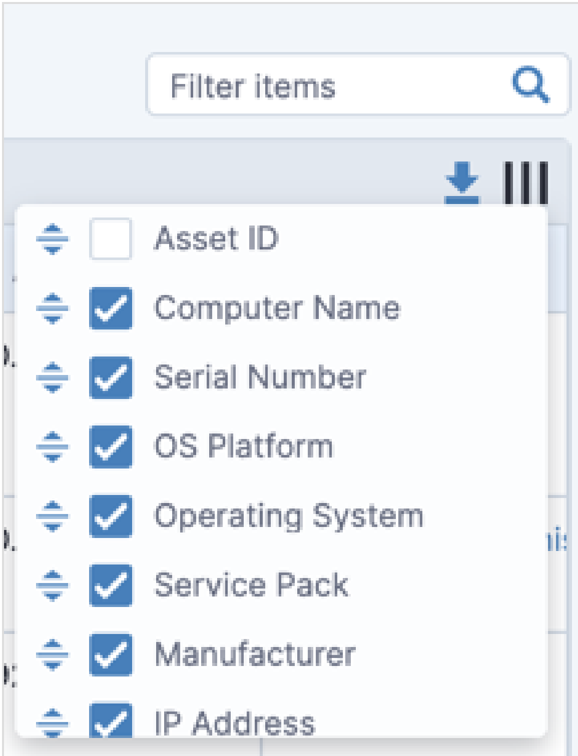
3. In the detail report, you can see more information about the selected asset.



Customize columns

You can change which columns are displayed in the report, and adjust the order of the columns.

1. In the report, click Customize Columns ||| .



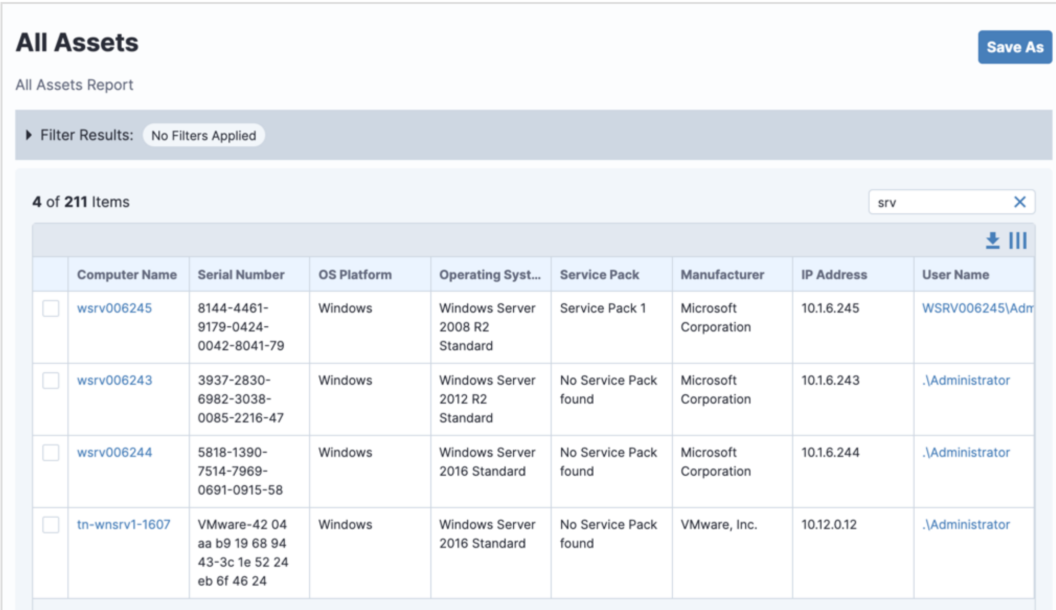
2. To remove a column, clear the checkbox for the column.
3. To adjust the column order, click and drag the column names.
4. Click **Save**.

Filter reports

To quickly search the values in the report, use filter items. To filter the values based on certain column values with operators (greater or less than, contains, is equal to, and so on), create a report filter.

FILTER ITEMS

In a report, you can quickly filter items by searching the report.



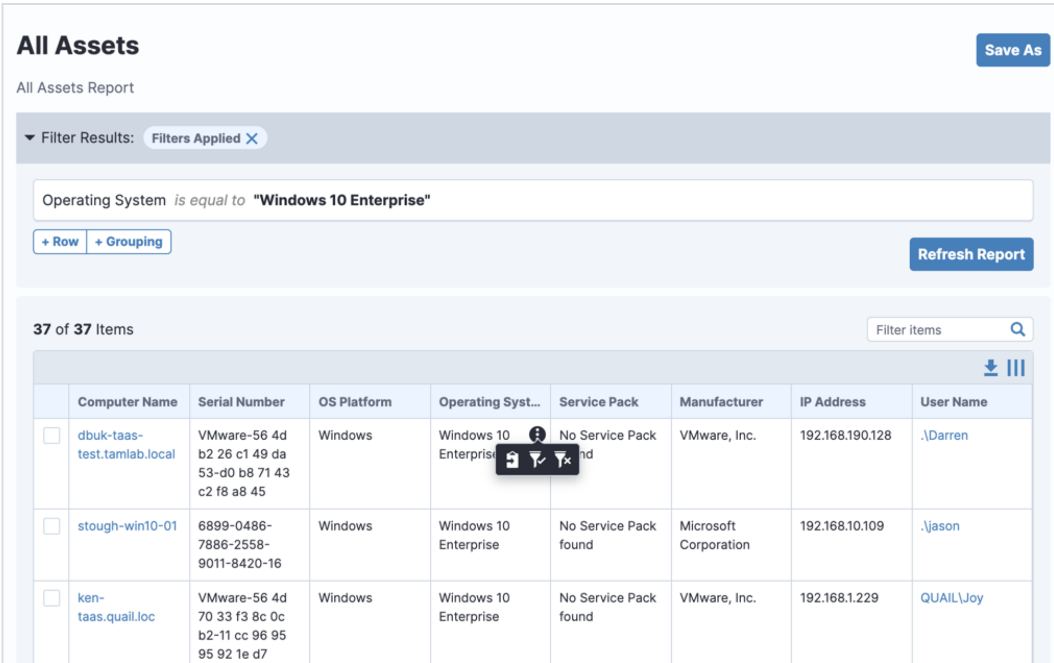
These filters are not saved when you leave the report page.

> **NOTE**
> You cannot filter on columns with numeric values.

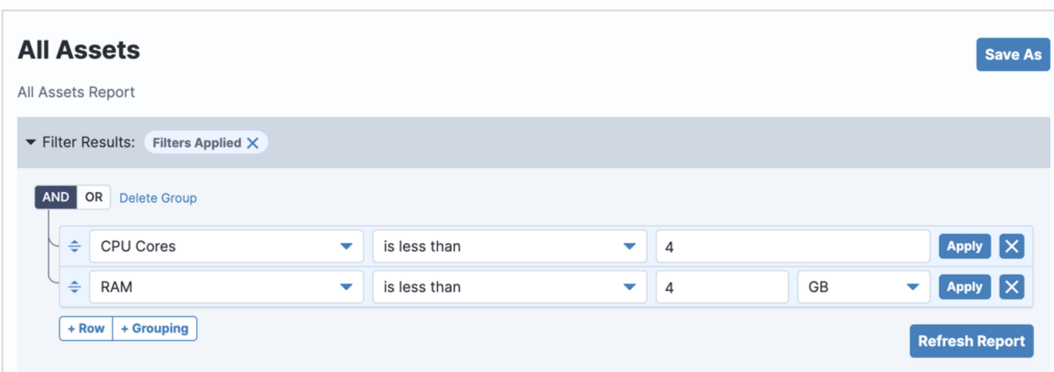
CREATE REPORT FILTER

You can perform live filtering on any report. You can also save the filter for future reference.

1. In a report, expand the **Filters** section. If the report contains a filter, that filter is already listed.
2. Add filters. You can add to the filter with the following methods:
    - Create a quick filter. Hover over a value in your report and click **Add is equal to filter** or **Add is not equal to filter**. The appropriate condition is added to the **Filter results:** section.



    - Edit the filter directly. Click **+ Row** to create a filter rule that is at the same level as the selected rule. When you create the rule, you can choose the attribute and whether the filter applies AND or OR operators with the other filters at that level. Click **Apply**. To create nested groups of filters, click **+ Group**.



3. Update the report data. Click **Refresh Report** to refresh the report based on the filters.

4. (Optional) Save report filters.

    - If you want to save the filter you defined in a custom report, click **Save**, or **Save As** to create a copy of the report that contains the filter.
    - If you want to save filtering for a predefined report, click **Save As** and create a copy of the report that contains the filter.

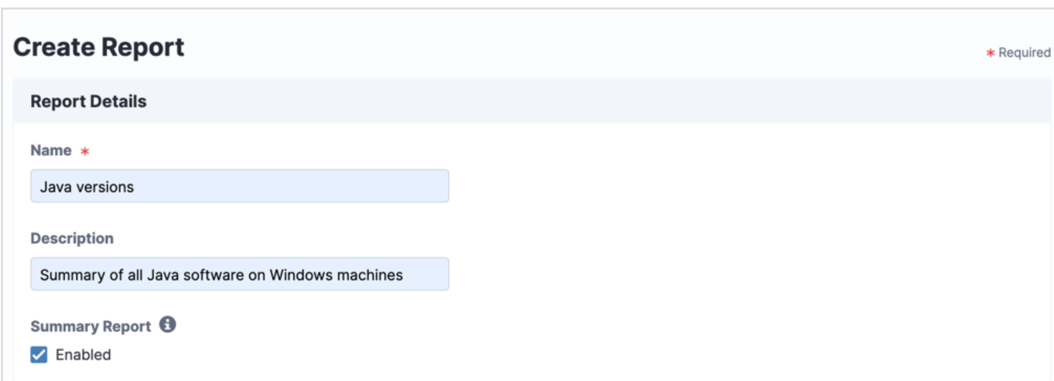
Create a report

You can create a new custom report, or you can copy from an existing report:

- To create a new report on the Reports page, click **Create Custom Report**.
- In an existing report, click **Save As** to create a copy of the report, give the report a new name, and click **Save**. The new report is displayed. Click **Edit** to change report details.

Specify general report information

1. Give your report a name and description to help you remember the purpose of the report later. The report name must be unique among all reports in Asset, including reports created by other users.
2. If you select **Summary Report**, the data is grouped into rows with an associated count that you can click to view more details.

    For example, if you create a report that lists SQL installations, with this option enabled, you get a result that lists a count of the computers that have each SQL version. You can then click the count to view the list of computers with that particular software installed.

Define asset filter

You can create a filter to specify attributes to identify the assets to include in the report. This filter is always applied when the report runs.

Click **+ Row** to create a filter rule that is at the same level as the selected rule. When you create the rule, you can choose the attribute and whether the filter applies AND or OR operators with the other filters at that level. Click **Apply**. To create nested groups of filters, click **+ Group**.

Select **Any** to include assets that match any specified attribute, or select **All** to include assets that match all specified attributes.

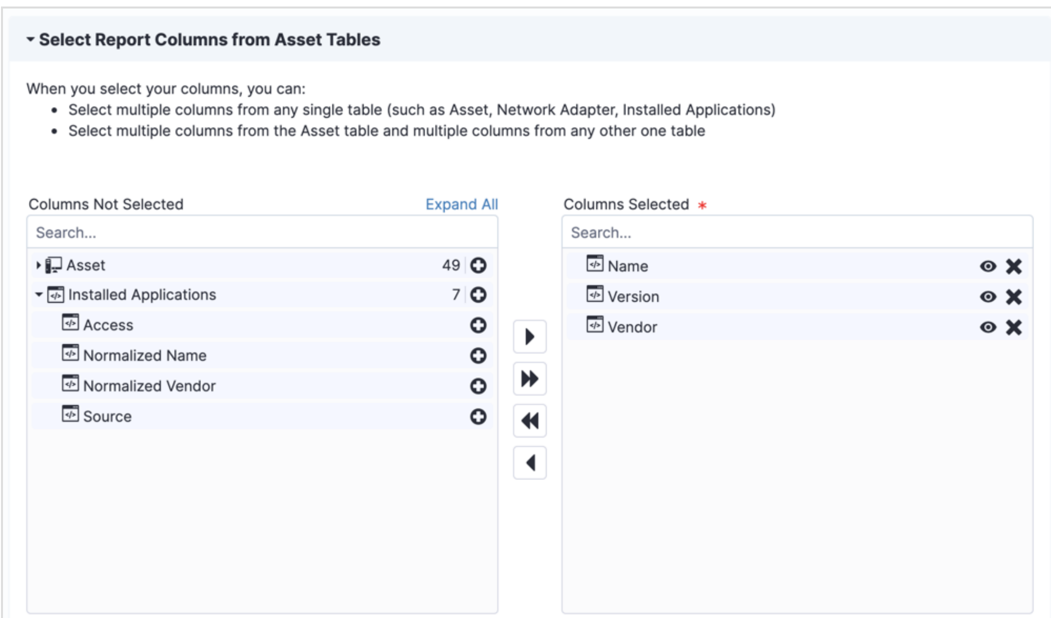
- To save a rule, click **Apply**.
- To edit a filter rule, click Edit.
- To discard changes to a rule you are editing, click Cancel.
- To delete a filter rule, click Delete.

Select columns

The columns that are available to include in your reports come from the asset sources that you define. To define sources, click **Inventory Management > Sources** in the Asset menu.

> TIP
>
> When you select columns, you can select multiple columns from the Asset table and any other one table (such as Network Adapter, Installed Applications, or Logical Disk). You can also select columns from any single table. For example, you cannot select columns from both the Network Adapter and Installed Applications tables.

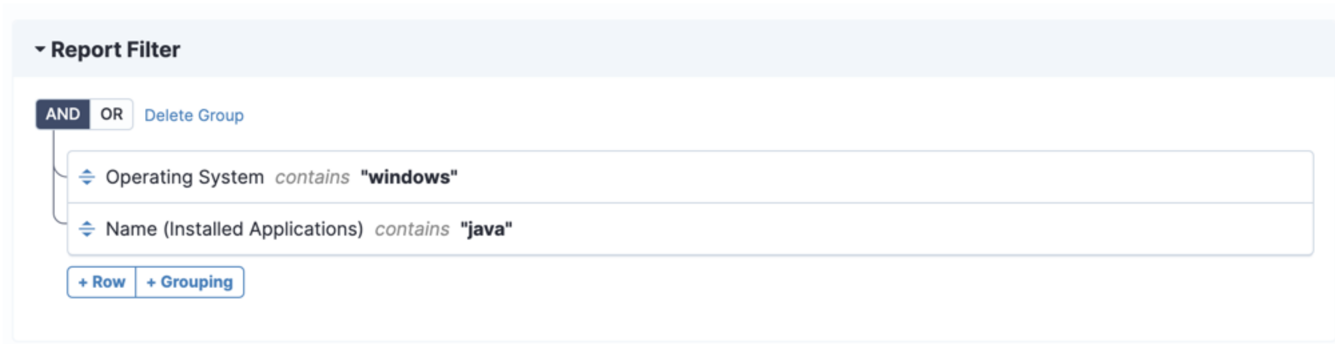
1. In the **Add Columns** section, select the data that you want to include in your report. You can search for the column that you want to use, or expand and collapse the data categories to find which columns you want to include.



2. Specify the order that you want your columns to display. In the **Columns Selected** section, click and drag the column names to arrange them in the order you want them from left to right in the resulting report. If you do not want the column to show up in the report but want the data to be available for filtering, change the **Show** value to **Hide**.

Define default filter

You can create a filter that is always applied when the report runs.



Click **+ Row** to create a filter rule that is at the same level as the selected rule. When you create the rule, you can choose the attribute and whether the filter applies AND or OR operators with the other filters at that level. Click **Apply**. To create nested groups of filters, click **+ Group**.

- To save a rule, click **Apply**.
- To edit a filter rule, click Edit.
- To discard changes to a rule you are editing, click Cancel.
- To delete a filter rule, click Delete.

Filter calculations

- If you use a `Size` data type in your report (such as the **Total Disk Space** or **Free Space** fields), the values are converted with a base of 1024. For example, 450,000 MB is converted to 439 GB.
- If you use a `Date` data type in your report (such as the **Created Date** field), you can filter on dates or calendar dates. For example, if you enter a filter for `created in the last "10 Days"`, the origin date and time are calculated from the exact current date and time. If you enter a filter `created in the last "10 Calendar Days"`, the origin date and time is always 12:00 AM on the specified number of days back from the current date. You can preview the date range when you save the rule.

Finish report

To save the report, click **Submit**. The report with the data is displayed. See Work with existing reports on page 36.

Delete report

To delete a report, go to the **Report** page and click Delete on the row of the report you want to delete. You can only delete custom reports.

Purge assets

Asset automatically purges stale assets from the database that have not been seen by Asset after the configured stale data age or based on a purge report. For more information, see Configure data retention and vacuuming on page 83.

You can manually remove assets from your asset database earlier than the stale data age. In a report that shows the assets you want to remove, select the rows and click **Delete**.

If the asset you deleted responds to the Asset saved questions in a subsequent load, the asset is re-added to the database.

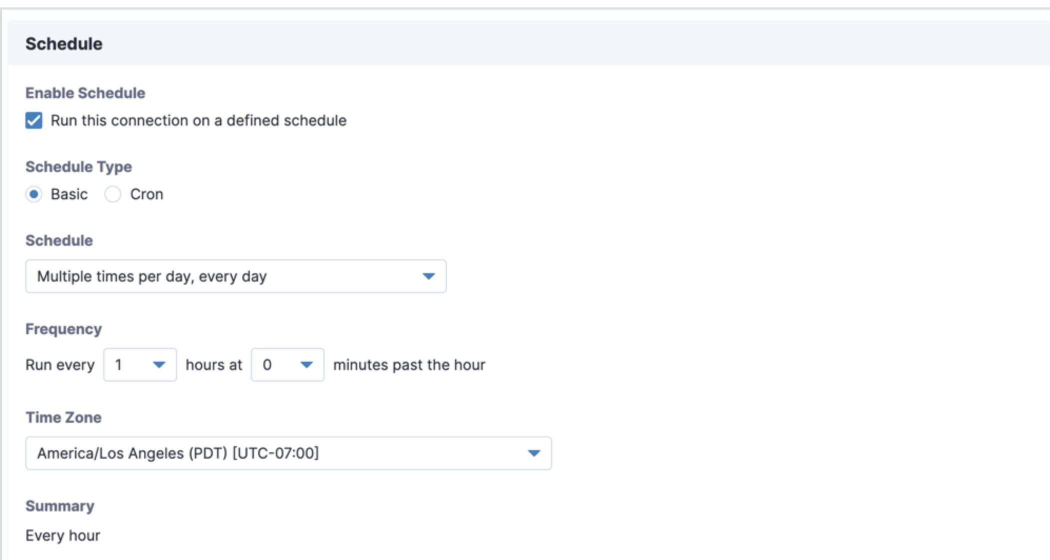
Configuring sources

Sources define how Asset gets data. By default, the source data in Asset comes from Tanium, populated from saved questions that are scheduled on a configurable basis. You can augment Asset inventory data with external data from a SQL Server database. By importing data that is typically not available on an endpoint into the Asset data store, you can enable filtering and reporting on information such as department, cost center, building, and location.

Configure Tanium source

The Tanium source is configured by default, and you cannot delete the Tanium source.

1. From the Asset menu, go to **Inventory Management > Sources**.
2. For the Tanium source, click **Edit** .
3. Configure the log level settings for the Asset import process.
4. Add the user name and password for the service account. The selected user must be a Tanium Administrator. This configuration is shared with the user name and password for the service account that is configured in the Settings on the **Service Account** tab.
5. Configure the schedule. The schedule determines how often asset data is imported from the Tanium live data into the asset database.



This database provides data for offline assets. You can create a standard interval or a Cron schedule. For example, you might create one of the following intervals based on your environment size:

- Less than 50,000 devices: every 1-2 hours.
- Less than 250,000 devices: every 4 hours.
- Greater than 500,000 devices: contact Tanium Support to configure the import schedule. For more information, see Contact Tanium Support on page 85.

For more information about the Cron syntax, see Reference: Cron syntax on page 86.

6. Click **Update** to save your changes.

What to do next

Add attributes from Tanium sensors into Asset. See Configuring attributes on page 49.

Configure database source

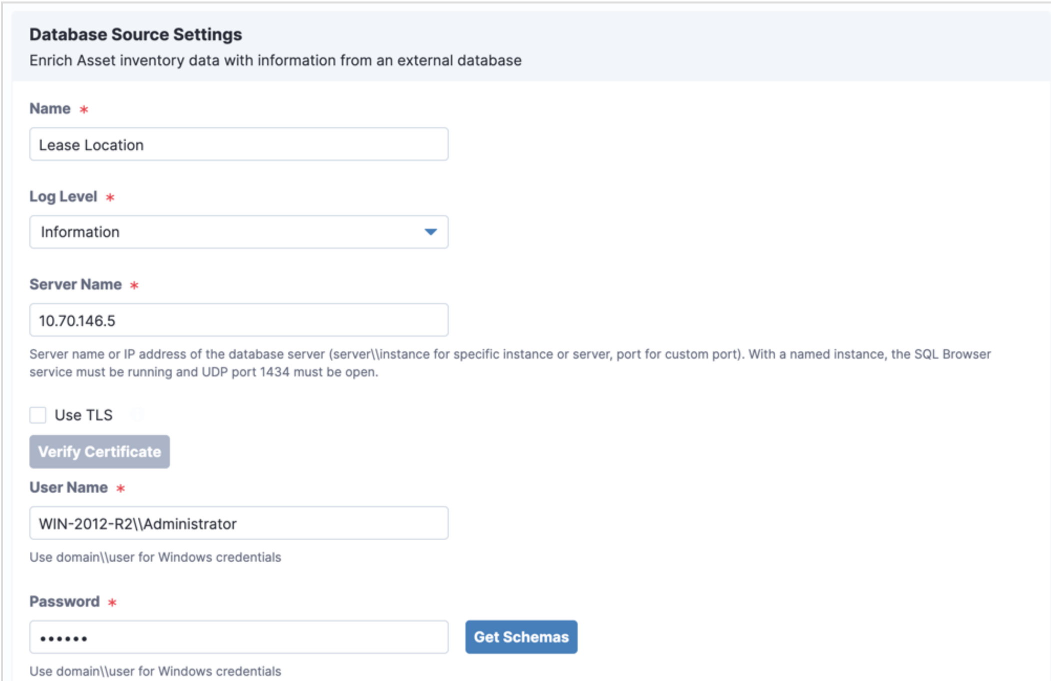
Enrich Asset inventory data with information from an external database. For example, you might have a database that has department or location information that you want to associate with the computers you are already tracking in Asset. To import this data, map columns from your SQL database to the Asset database.

When the data import runs, each row of your import table is evaluated against the data that is already in Asset. For example, if you create a mapping for your computer ID in the import table, that ID gets matched to the Computer ID column in Asset. If a match is found, the attributes that you configure to get imported from your source table are inserted into Asset. If you map an import column that has duplicate values in the table, only the last value is stored in Asset after the load has completed.

IMPORTANT: If you use column types (like `nchar`) that pad values with spaces, your identity mappings on the database source might not work correctly. For example, a computer name like `win1` might come through as `win1      ` (with six spaces after it). These values do not match a value `win1` in Asset.

1. From the Asset menu, go to **Inventory Management > Sources > Create Database Source**.
2. Configure source settings. Enter a name for the data source and the server name.
3. (Optional) If your database server has Transport Layer Security (TLS) configured, select **Use TLS** and click **Verify Certificate**. To establish trust on first use (TOFU), review the certificate details and click **Verify**.
   For more information about enabling TLS on SQL Server, see Microsoft SQL Docs: Enable Encrypted Connections to the Database Engine.
4. Enter the username and password for the database connection.

5. Choose the database and schema. Click **Get Schemas**. Asset connects to the database and gets the database and schema information.



6. Configure source mappings. Source mappings uniquely identify the assets in your source data, and define how the columns in your source table relate to the Asset database. For an attribute to be available as a reference for mapping, it must only return a single row of value. You can add multiple source mappings from your database source. All the mappings must match for the data to be imported into the Asset database.

   a. Click **Add Mapping**. Choose a source table.

   b. Create identification rules. The columns associated with the selected source table are loaded into the **Source Columns** field.

   c. Choose a destination attribute. Choose a column in the Asset database. You can create multiple identification rules as necessary.

7. Configure the import schedule. The schedule determines how often asset data is imported from the external database into the asset database. You can create a standard interval or a Cron schedule.
   For more information about the Cron syntax, see Reference: Cron syntax on page 86.

8. Click **Create**.

If you want to modify the source, go to **Inventory Management > Sources**. Click Edit. You must enter the credentials for your SQL server again before you can modify the source mappings.

> IMPORTANT
> A maximum of 20,000 asset updates are performed for each 100 database source records during the load process. You might need to run an import several times to update all of the data.

What to do next

Add the attributes from your import table into Asset. See Configuring attributes on page 49.

Disable sources

Data imports for a source run on a configured schedule by default. You can disable this schedule to resolve issues, or run the import manually.

1. From the Asset menu, go to **Inventory Management > Schedules**.
2. Hover over the import schedule for the source that you want to disable and click Edit.
3. Clear the **Run this connection on a defined schedule** setting. Click **Update**.
4. If you want to manually run an import, go to the **Inventory Management > Schedules** page. Hover over the import schedule and click **Run Now**.

Configuring attributes

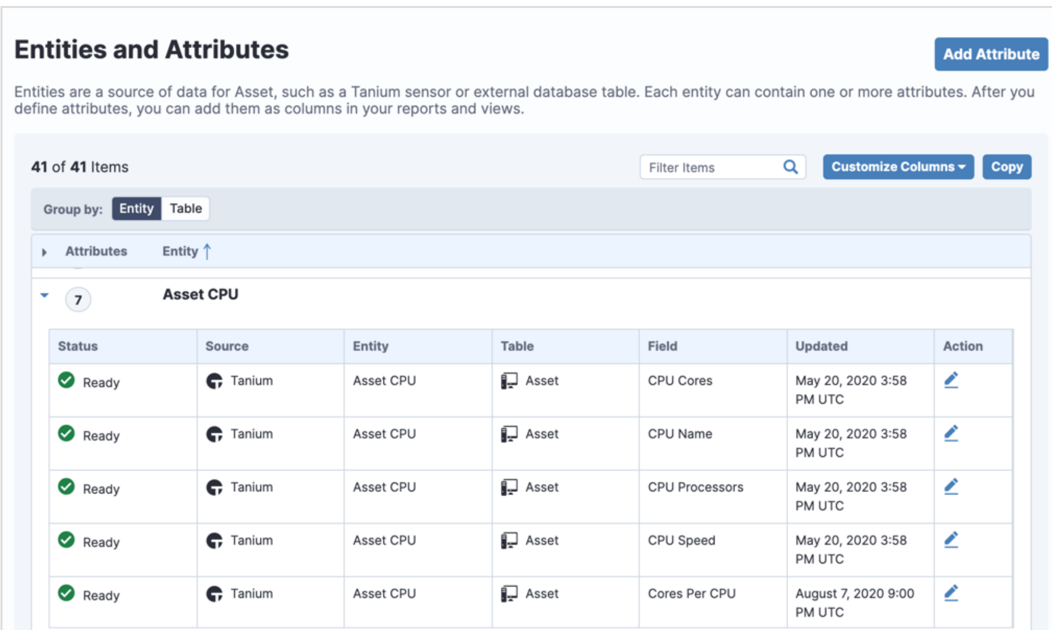
You can configure the attributes that get populated as asset data. These attributes can come from Tanium or a configured source entity. These attributes become the columns that you can include in your reports.

Asset solution content

When Asset is installed, solution content is also installed. Some of this solution content is similar to the content packs that got imported when you installed Tanium. In most cases, the Asset version of the content adds attributes or improves the information beyond the existing sensor. For example, the **Asset CPU** sensor includes columns for speed, cores, and processors, but the **CPU** sensor from the **Core Content** content pack contains one column. The Asset custom content is hidden on the **Administration > Content > Sensors** page.

View and edit default asset attributes

By default, Asset imports about 70 common attributes from Tanium sensors. To view the list of default attributes, go to the Asset menu and click **Inventory Management > Attributes**.



- You can group by the table in the Asset database, or the entity type (for Tanium attributes, the entity types match the sensor names).
- To sort a column, click the column header.
- To filter the list on any field, use the **Filter Items** search.

- To customize the columns that are displayed and column order, click **Customize Columns**. For example, to view the Column name returned from a sensor, select **Source Attribute**. You can add or remove columns, and then click and drag the columns to an order that you want to display.
- To copy a CSV version of the data listed in the table to the clipboard, click **Copy**.
- To edit an individual attribute, click Edit.

Normalized attributes

The **Installed Applications** table contains fields for **Normalized Name** and **Normalized Vendor**. Normalized values resolve variants on a vendor or software name. For example, the following vendor values: `CompanyName, Inc.`, `CompanyName Inc`, and `CompanyName LLC`, are resolved to `CompanyName` for the **Normalized Vendor** column.

Normalization occurs after data load. If values are in the **Name** or **Vendor** columns, a normalized value is also calculated.

Configure additional Tanium attributes

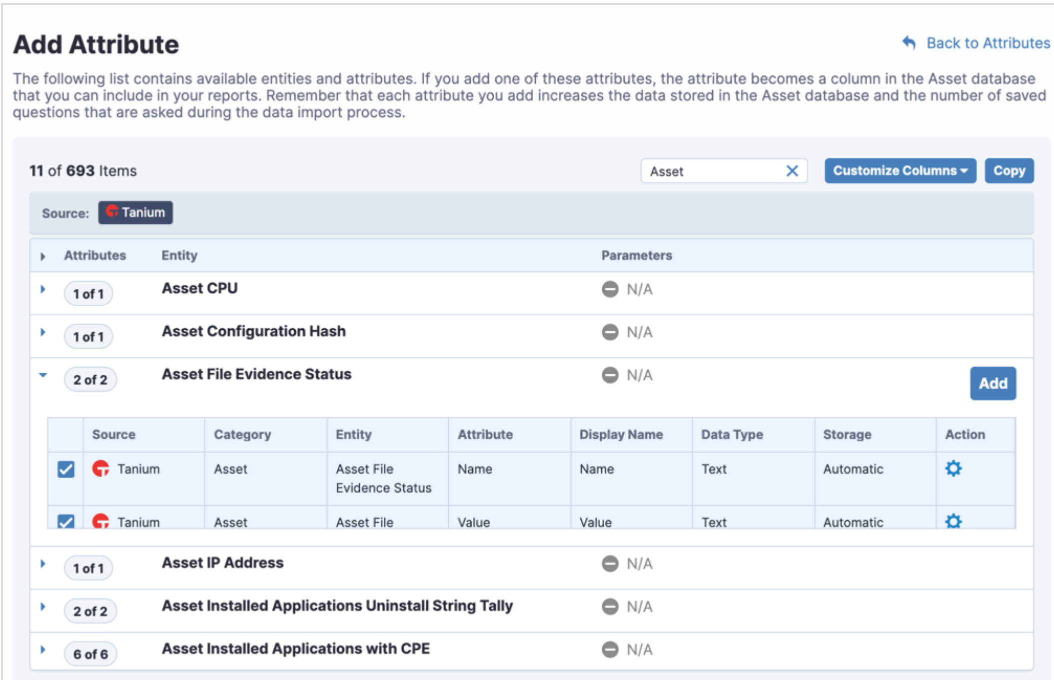
You can import attributes from Tanium sensor data. For example, you might choose to import other critical configuration information from other Tanium products, solutions, or custom content.

Important considerations for attributes

- Remember that each attribute you add increases the number of saved questions that are asked during the data import process. Each new attribute also adds columns and data to the Asset database.
- Add sensors that return a single line when possible.
- Before you add a new attribute, determine the storage method. Using the automatic evaluation of storage method sometimes results in a single-row sensor being seen as a multiple-row sensor, which is less efficient and can have unforeseen connotations when reporting.
- Avoid adding attributes that change often, especially attributes that return a large amount of data.
- Consider how frequently Asset inventories your environment. If the attribute data becomes stale before the next schedule import, it might not be a good attribute to add.
- Gather data that you need more frequently than the scheduled import process with Interact or a saved question.

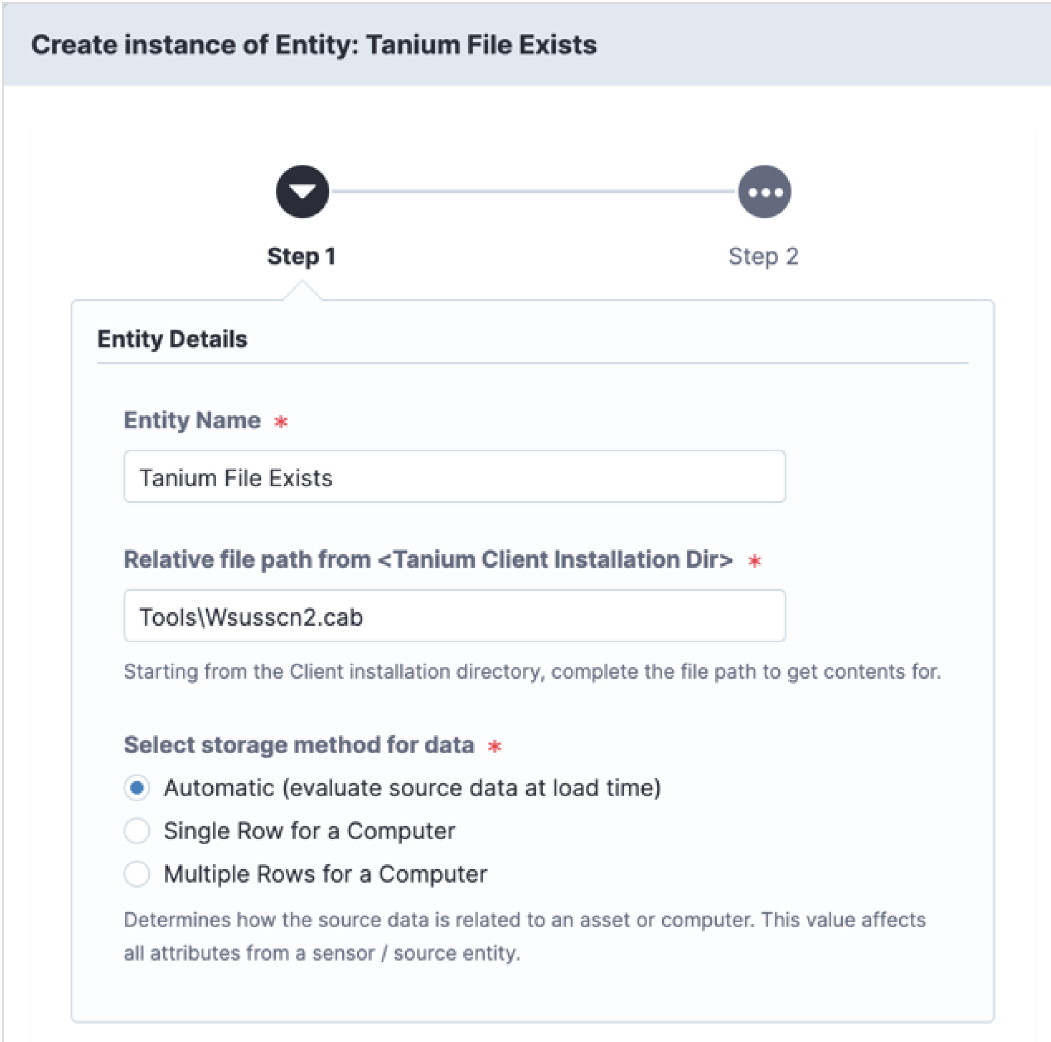
Add attributes

1. From the Asset menu, click **Inventory Management > Attributes**. Click **Add Attribute**.
2. Search for and select the sensors from which you want to add attributes. To view all of the Asset custom content, you can search for `Asset` in this sensor list.

   Select the attributes that you want to import. Click **Add**.
3. (Optional) To change the data type, display name, or storage method for a selected attribute, click Settings ⚙ for the associated attribute.

4. If you are adding an attribute from a parameterized sensor, indicate the values that you want to pass to the parameters.



You can also choose how to store the data that gets returned in the database. If you set the storage method for data to be automatically evaluated at load time, this selection adds time to the load process. Click **Next**.

5.

Click **Create** and verify that the changes are correct.

> IMPORTANT
> This step makes changes to the Asset database. Review and verify carefully.

6. After the database updates are completed, you can see the new attribute in the main list of attributes in **Pending** state. The new attribute stays in pending state until the next time the Tanium source import job runs, and data gets populated in the database. To run this job immediately, see View schedule and run import on page 54. When populated, the attribute displays in **Ready** state. You can add attributes that are in **Ready** state to custom reports.

Configure external attributes

You can also import attributes from an external data source.

Before you begin

If you are importing data into Tanium from an external data source, configure the import first. Configuring the source defines which database tables to import and how to correlate the data with the ci_item table in the Asset database. See Configuring sources on page 45.

Add external attributes

1. From the Asset menu, click **Inventory Management > Attributes**. Click **Add Attribute**.
2. In the **Source** column, find the data source that you configured. You can search for and select the field that you want to add as an attribute.
3. Asset automatically converts values in the database based on the following type mapping:

SQL data type	Asset data type
BIGINT	BIGINT
DATE	DATEONLY
DATETIME, TIMESTAMP WITH TIME ZONE	DATE
BIT, BOOLEAN, TINYINT	BOOLEAN
STRING, CHARACTER VARYING (255), VARCHAR (255)	STRING
FLOAT, DOUBLE PRECISION	FLOAT
INT, INTEGER, SMALLINT	INTEGER
CHAR, NCHAR, NVARCHAR, TEXT, VARCHAR	TEXT

4. Click **Create** and verify that the changes are correct.

   IMPORTANT: This step makes changes to the Asset database. Review and verify carefully.

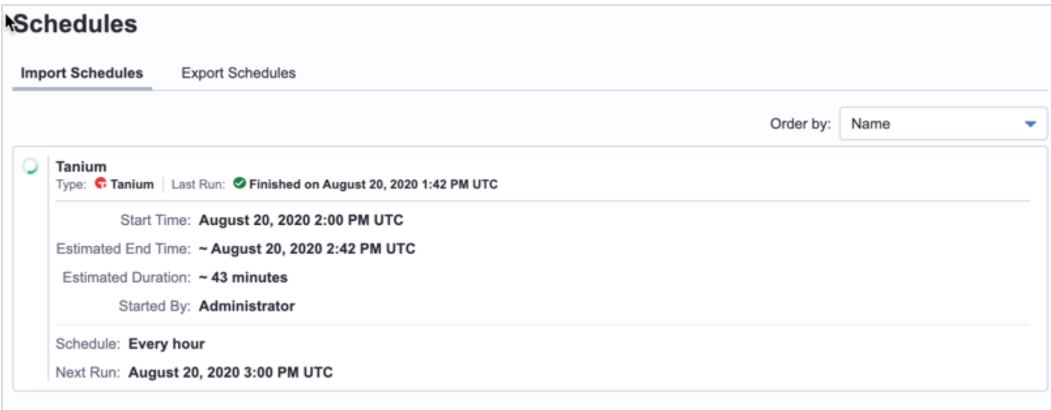
5. After the database updates are completed, you can see the new attribute in the main list of attributes in **Pending** state. The new attribute stays in pending state until the next time the **External database** job runs, and data gets populated in the database. To run this job immediately, see View schedule and run import on page 54. When populated, the attribute displays in **Ready** state. You can add attributes that are in **Ready** state to custom reports.

Schedule and run Asset data imports

The import asset data schedule determines how often Asset pulls data from Tanium or an external data source to save in the Asset database. This database is used to provide information for offline assets.

View schedule and run import

1. From the Asset menu, click **Inventory Management > Schedules**. On the **Import Schedules** tab, the schedule is displayed.



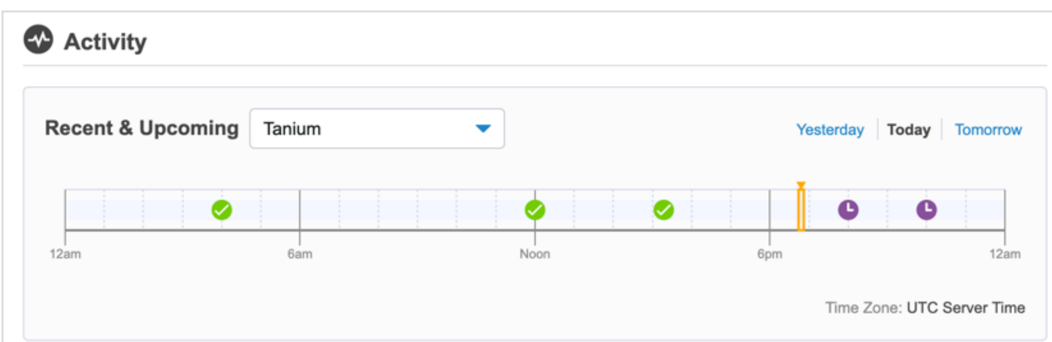
2. Hover over the schedule and click **Run Now** to override the schedule and have Asset immediately pull source data into the Asset database.

3. If you have any attributes that are in **Pending** state, you can watch the attributes change to **Ready** state on the **Inventory Management > Attributes** page. You can add attributes that are in **Ready** state to custom reports.

If you want to change the import schedule, update the asset source. See Configuring sources on page 45.

View imports

On the Asset **Overview** page in the Activity section, you can view a timeline of the recent imports and exports.



The timeline contains each import, with one of the following statuses:

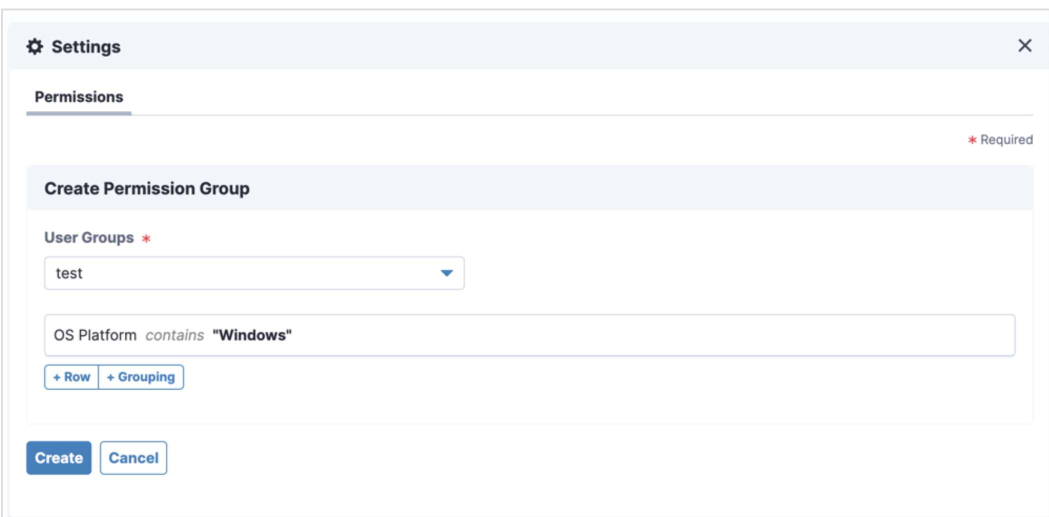
- : Scheduled
- : Successful
- : Error

Set user permissions on computers

In addition to the Asset user roles that control access to Asset reports and settings as a whole, you can define more detailed permissions on computers that are based on individual attribute values. For example, you can create permissions that assign a user group permission to access information about Windows or macOS platform assets only. If a user belongs to multiple user groups, the permissions for all the user groups are combined with an OR operator.

Before you begin, you must have a user group to which you want to assign the Asset permissions. See Tanium Core Platform User Guide: Managing user groups. For the users in this user group to access Asset, they also must have an Asset user role assigned. See User role requirements on page 25.

1. From the Asset **Overview** page, click Settings . Click the **Permissions** tab.
2. Click **Create Permissions**.
3. Select the user group from the list that you want to assign.
4. Add a condition. This list of attributes is from the **ci_item** table of the Asset database.
   For example, to assign the user group permission to view Windows assets only, set to `OS Platform contains Windows`.

5. Click **Apply** to create each condition.
6. Click **Create**.

Delete attributes

To delete an attribute, you must disable the attribute, then remove references.

1. Disable the attribute.
    a. From the Asset menu, click **Inventory Management > Attributes**.
    b. Click Edit next to the attribute that you want to disable.
    c. Clear the **Enabled** setting. Click **Save**.
2. Remove references to the attribute from all reports, views, permissions, and ServiceNow destinations.
    a. Click Delete next to the attribute that you want to remove.
    b. A list of known locations where the attribute is used is displayed. Click the link for each item, and edit the item to remove the use of the attribute.
    c. Click **Refresh** to list the remaining references.
3. After you remove the references, type the name of the attribute to confirm its deletion. Click **Delete Attribute**.

The attribute is deleted from the Asset database on the next load of data from the data source.

Configuring views

Use views to create filtered versions of Asset data for exporting to destinations. Depending on your destination, you might need to create views or Asset might automatically create views for you.

Destination	How view is created
Tanium Connect	Create custom views on page 57
Flexera FlexNet Manager Suite 2019 R2.2 or later	Create Flexera views on page 59.
Flexera FlexNet Manager Suite 2019 R1 or earlier	(Automatic) Asset creates views when you create the destination.
ServiceNow: Tanium Asset integration	(Automatic) Asset creates a view when you create the destination.
ServiceNow: Service Graph Connector	Create ServiceNow view on page 59.

Create custom views

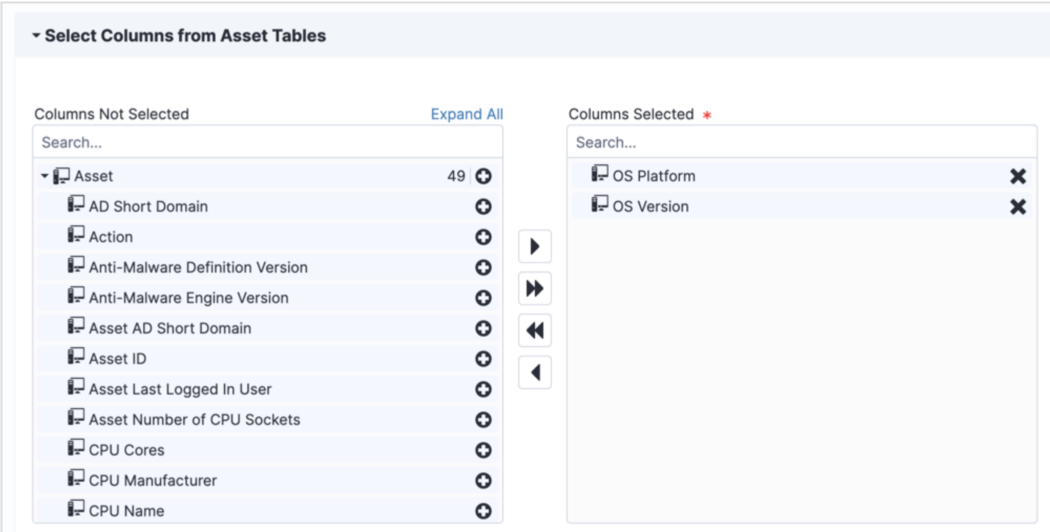
You can specify attributes to include in the data set and use filters to limit the computers that are included in the view. You can include fewer attributes than the default view, depending on the use of your view.

Specify view details

1. From the Asset menu, click **Views**. Click **Create View**.
2. Give your view a name and description to help you remember the purpose of the view later. The view name must be unique among all views in Asset, including views created by other users.

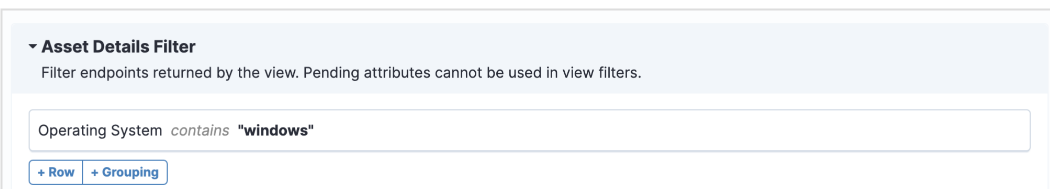
Select columns

In the **Select Columns from Asset Tables** section, select the data that you want to include in your view. You can search for the column that you want to use, or expand and collapse the data categories to find which columns you want to include.



Define filter

(Optional) Add filters to limit the computers that are included in the view. You can create filters for Asset Details, Asset Installed Applications, Asset Logical Disk, Asset Network Adapter, Asset Physical Disk, Asset Windows Installer Applications, or any custom reference attributes that you have added to Asset. The available filters correspond to the data categories for the selected columns.



Click **+ Row** to create a filter rule that is at the same level as the selected rule. When you create the rule, you can choose the attribute and whether the filter applies AND or OR operators with the other filters at that level. Click **Apply**. To create nested groups of filters, click **+ Group**.

Click Expand to view a JSON representation of the rule, which can be helpful to evaluate complex filtering.

- To save a rule, click **Apply**.
- To edit a filter rule, click Edit .
- To discard changes to a rule you are editing, click Cancel .

- To delete a filter rule, click Delete .

By default, computers are in the view even if the fields in the filter do not have associated values. If you want to require that results are returned for the fields in the filter for a row be included in the view, select **Should be filtered out of the view** for the filter.

Filter calculations

- If you use a `Size` data type in your report (such as the **Total Disk Space** or **Free Space** fields), the values are converted with a base of 1024. For example, 450,000 MB is converted to 439 GB.
- If you use a `Date` data type in your report (such as the **Created Date** field), you can filter on dates or calendar dates. For example, if you enter a filter for `created in the last "10 Days"`, the origin date and time are calculated from the exact current date and time. If you enter a filter `created in the last "10 Calendar Days"`, the origin date and time is always 12:00 AM on the specified number of days back from the current date. You can preview the date range when you save the rule.

Finish and test the view

1. To save the view, click **Create**.
2. Test the view. On the **Views** page, hover over a view and click Test View , which runs the view for a single asset. The resulting message states whether the view can run successfully, or contains errors such as pending attributes.

Create ServiceNow view

If you are using ServiceNow Orlando Patch 7 or later, you must create a ServiceNow view. The view creates the ServiceNow-specific data that is exported from Tanium into ServiceNow.

> NOTE
> You do not need to create a ServiceNow view if you are using ServiceNow: Tanium Asset integration. The view is automatically created when you create the ServiceNow destination.

1. From the Asset menu, click **Views**. Click **Create ServiceNow View**.
2. Test the view. On the **Views** page, hover over a view and click Test View , which runs the view for a single asset. The resulting message states whether the view can run successfully, or contains errors such as pending attributes.

Create Flexera views

If you are using the Flexera FlexNet Manager Suite 2019 R2.2 or later, create the Flexera views. The views create the Flexera-specific data that is exported from Tanium into FlexNet Manager Suite.

> NOTE
> You do not need to create Flexera views if you use FlexNet Manager Suite 2019 R1 or earlier. The views are automatically created when you create the Flexera destination.

1. From the Asset menu, click **Views**. Click **Create Flexera Views**.
2. Test the view. On the **Views** page, hover over a view and click Test View, which runs the view for a single asset. The resulting message states whether the view can run successfully, or contains errors such as pending attributes.

Export data from views

To see what data is in a view, you must create a destination.

TIP

Using the Tanium Asset API, you can also view the data with any tool that allows you to make and inspect web requests without creating a destination.

Tanium Connect

Export data from a view to Connect destinations such as Email, File, HTTP, Socket Receiver, Splunk, and SQL Server. See Tanium Connect on page 64 for more information about configuring connections and the recommended formatting options to use for each connect destination type.

ServiceNow

The ServiceNow view contains default fields and filters for integration with ServiceNow.

This reserved view is read-only. If you want to customize the fields or filters, click **Create Copy**. Edit and rename the view. You might want to create a new view to limit the computers that are included in the ServiceNow export or to include new attributes in the data, in conjunction with modifying the mappings.

For more information, see ServiceNow: Tanium Asset integration on page 72.

Flexera

The Flexera views contain default fields and filters for integration with Flexera.

These reserved views are read-only. If you want to customize the fields or filters, click **Create Copy**. Edit and rename the view. You might want to create a new view to limit the computers that are included in the Flexera export.

For more information, see Flexera FlexNet Manager Suite: 2019 R2.2 or later on page 67 or Flexera FlexNet Manager Suite: 2019 R1 or earlier on page 68

Monitoring software inventory and usage

You can track and monitor software inventory and usage on endpoints. For example, you might track usage of a product to confirm that your enterprise is using the number of purchased licenses for the product and reclaim unused licenses.

The process for monitoring software inventory usage involves both collecting and reporting data.

1. After the Tanium Client is installed on a Microsoft Windows or macOS endpoint, Asset starts collecting software inventory and usage on the endpoint. Usage is collected each minute.

2. Reports on software inventory and usage generate after the following conditions are met:

   - You track products to enable reporting in the Asset **Software Inventory & Usage > All Products** page.

   - The Tanium Client and tracked product are installed for at least seven days or the specified baseline period.

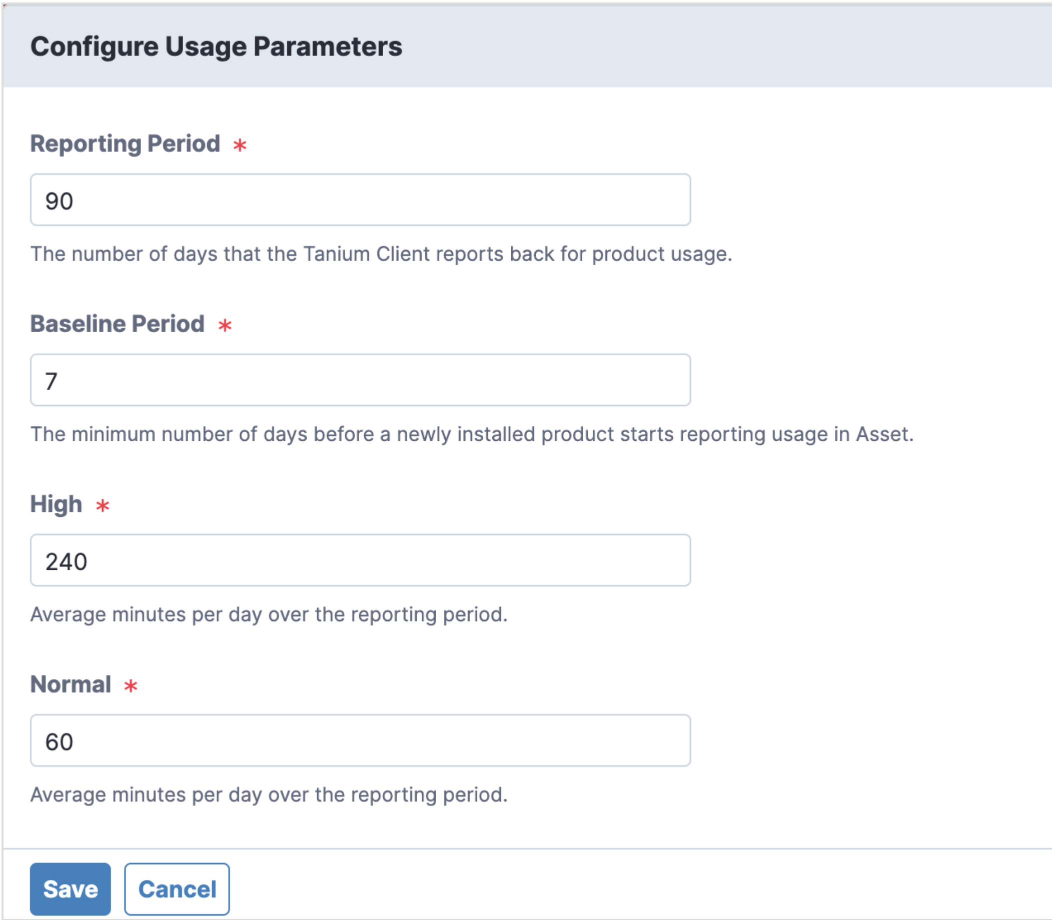
   Asset reports the vendor, product, version, number of endpoints with installed product, and usage of the installed product on the endpoints, including last used date and usage levels.

TIP

If you want to view software usage for products that are not tracked and reported in Asset, use the **SIU - Installed Products, SIU - Product First Used, SIU - Product Last Used,** and **SIU - Product Used** sensors. For assistance, see Contact Tanium Support on page 85.

Track a product to enable reporting

You must track a product to enable reporting in Asset. You can only track a product or suite listed on the **Software Inventory & Usage > All Products** page. For example, if Microsoft Office 365 is listed on the page, Asset tracks usage for the suite and not for individual products within the suite.

1. From the Asset menu, click **Software Inventory & Usage > All Products**.
2. Click Start Tracking next to the appropriate product.
3. 

   To edit the usage parameters, in the **Action** column, click Edit Parameters .

   a. Configure the reporting and baseline periods.

   b. Specify the total number of minutes per day for high and normal usage levels. The minutes do not need to be consecutive. The default high value is 240. The default normal value is 60.

   c. Click **Save**.

4. To stop tracking usage for a product, click Stop Tracking .

View software inventory and usage

You can view software inventory and usage details from the **Software Inventory & Usage** and **Reports** menus. To drill down for specific endpoint details, use the reports available in the **Reports** menu.

View product inventory and general usage

1. From the Asset menu, click **Software Inventory & Usage**.
2. Click **All Products** or **All Vendors** to review the data.

> NOTE
> **All Products** only displays data for tracked products.

3. To export the data to a CSV file, click Export. The export includes the data as it is currently displayed in the data grid.

View last used date of a product

The last used date is the date the endpoint last reported usage for the product. For endpoints shared with multiple users, this is the last date that any user on the endpoint used the product.

1. From the Asset menu, click **Reports > SIU Product Last Used**.
2. View the **Last Used Date** for the product.

View usage levels of a product

1. From the Asset menu, click **Reports > SIU Product Usage**.
2. View the **Usage** for the product.

The report includes the average daily usage level for a product.

Level	Description
High	Number of minutes defined in **Software Inventory & Usage > All Products > Edit Parameters**.
Normal	Number of minutes defined in **Software Inventory & Usage > All Products > Edit Parameters**.
Limited	Number of minutes between one and the Normal usage level.
Baselining	Product is newly installed and not reporting usage in Asset. The baseline length is defined in **Software Inventory & Usage > All Products > Edit Parameters**.
Usage not detected	Tanium Client does not detect usage of the product. For example, the baseline period has passed and the product has not been used.
Not installed	Endpoint does not have the software installed or is unable to install the software.

View installed version of product

1. From the Asset menu, click **Reports > SIU Installed Products**.
2. View the **Version** for the product.

Exporting data to destinations

You can configure Asset to export data to external destinations, such as Tanium Connect, ServiceNow, and Flexera.

CSV file

You can copy data from an asset report table to the clipboard and paste the data in an application that can interpret a CSV file, such as a database. Select the rows that you want to copy and click **Copy**. You can then paste the information about the rows you selected.

To save the entire report as displayed in a CSV file, click Export. You might see an **Export requested** status message before you see a **Download** link for the CSV file. Each Asset user can export a single report at a time. Multiple Asset users can be signed in, each exporting a single report at the same time.

For scheduling, formatting, and large data sets, consider setting up a Connect destination.

Tanium Connect

To export data from Asset to Connect destinations such as Email, File, HTTP, Socket Receiver, Splunk, and SQL Server, create a connection.

IMPORTANT SQL Server on Azure, Amazon Relational Database Service, and other database as a service providers are not supported as **SQL Server** destinations.

Before you begin

- You must have access to Connect with **Connect User** role.
- You must have an Asset report or view from which you want to export data. See Building reports on page 36 and Configuring views on page 57.

Create a connection

With Connect 4.3 to 4.7, choose **Asset Report** as the connection source. You can choose a predefined or custom Asset report as a connection source.

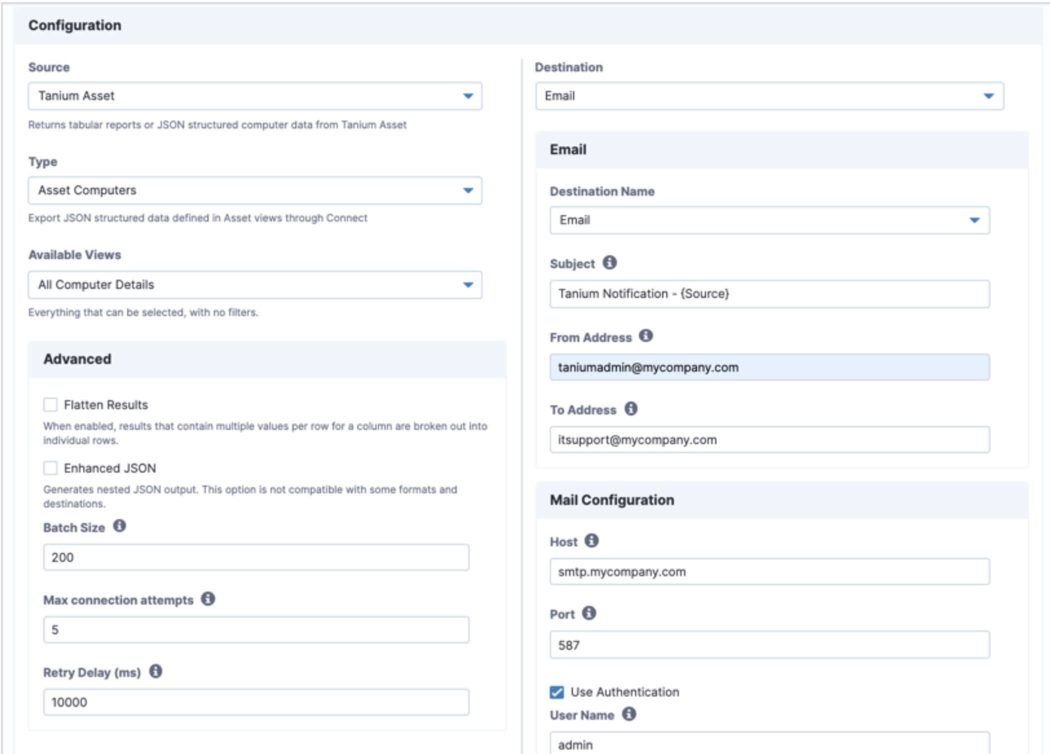
With Connect 4.8 and later, choose **Tanium Asset** as the connection source. You can choose from the following types of Asset source:

ASSET REPORTS

Select any predefined or custom report as the connection source.

ASSET COMPUTERS

1. Select any view as a connection source and export structured data using an Asset view.



2. If you do not enable **Flatten Results**, the entire data set that is retrieved for one computer is a single record. For example, if you are exporting Installed Applications, a computer has a single row with the entire list of installed applications in that same record. Any change that is made to this data set shows up in the destination. By enabling the **Flatten Results** setting, each installed application for a computer is processed as a single record.

3. With **Enhanced JSON**, the results contain an array of objects for each reference table, instead of an array of strings, numbers, or dates for each reference attribute. The correlation between attributes and destinations can be easier to implement.

4. Use the default batch size (200). Higher batch sizes can be slower due to increased processing for each batch.

5. Configure the export format.

   - If you enable **Enhanced JSON**, you must also choose JSON as the format for your connection.
   - To customize the column names, expand the **Columns** section and click **Add or Modify Columns**. Change the display values in the **Destination** column as needed.
   - If you customize the columns, leave the **Value Type** as **Unmodified** to get the expected object output.

COMPATIBILITY

Use the following recommendations for Enhanced JSON and Flatten Results settings for each format in Connect. If you use an unsupported combination, connection failures might occur or incorrect data might get written to the destination.

Connect destination format compatibility

Format	Enhanced JSON Support	Recommendation
CEF	❌	Use Flatten Results
CSV	❌	Use Flatten Results
Delimiter separated	❌	Use Flatten Results
Elasticsearch	✅	Use Enhanced JSON without Flatten Results
HTML	✅	Use Flatten Results without Enhanced JSON
JSON	✅	Use Enhanced JSON without Flatten Results
LEEF	❌	(Optional) Use Flatten Results
SQL Server	❌	(Required) Use Flatten Results
Syslog	❌	(Optional) Use Flatten Results

FILTERING `NO RESULTS` AND ERRORS

Asset automatically filters `no results` and common error messages and does not send that data to the connection destination.

EXAMPLES

Compare the data that is returned from Asset installed applications. View JSON examples

EXAMPLE: ENHANCED JSON

```
{
        "Computer Name": "WIN-2012-R2",
        "Asset Installed Applications": [{
                "name": "Tanium Server 7.2.314.3019",
                "version": "7.2.314.3019",
                "vendor": "Tanium Inc."
        }, {
                "name": "Microsoft Visual C++ 2015 Redistributable (x64) - 14.0.23026",
                "version": "14.0.23026.0",
                "vendor": "Microsoft Corporation"
        }, {
                "name": "Microsoft SQL Server 2012 Native Client",
                "version": "11.3.6540.0",
                "vendor": "Microsoft Corporation"
                }]
}
```

EXAMPLE: FLATTENED JSON


```
[{
        "computer name": "WIN-2012-R2",
        "name": "Tanium Server 7.2.314.3019",
        "vendor": "Tanium Inc.",
        "version": "7.2.314.3019"
}, {
        "computer name": "WIN-2012-R2",
        "name": "Microsoft Visual C++ 2015 Redistributable (x64) - 14.0.23026",
        "vendor": "Microsoft Corporation",
        "version": "14.0.23026.0"
}, {
        "computer name": "WIN-2012-R2",
        "name": "Microsoft SQL Server 2012 Native Client",
        "vendor": "Microsoft Corporation",
        "version": "11.3.6540.0"
}]
```


For more information about creating connections, see Tanium Connect User Guide.

Flexera FlexNet Manager Suite: 2019 R2.2 or later

If you have FlexNet Manager Suite 2019 R2.2 or later, you can use the Tanium Connector in FlexNet Manager Suite (FNMS) and the existing Tanium Client on your endpoints to export inventory data from Tanium into FNMS. Asset includes content with sensors that are specific to Flexera, including MS Exchange server, SQL server, Last Logged In, Number of CPU sockets, Short Domain, and so on.

FlexNet Manager Suite requirements

- FlexNet Manager Suite 2019 R2.2 or later
- FlexNet Beacon 14.2 or later

Tanium requirements

- Tanium Server 7.3.0 or later
- Tanium Asset 1.11 or later
- A user with the Asset Report Reader role to use with the Tanium Asset API

IMPORTANT You do not need to configure a Flexera destination in Asset. If you already have a Flexera destination configured in Asset that you set up for a previous version of Flexera, disable the destination after you complete the integration instructions.

Integration instructions

For detailed integration prerequisites and instructions, see "Tanium Connector" in [Flexera FlexNet Manager Suite Inventory Adapters and Connectors Reference].

> **NOTE** To create the Flexera views, from the Asset menu, click **Views > Create Flexera Views**. Do not edit or delete these views.

Flexera FlexNet Manager Suite: 2019 R1 or earlier

If you have Flexera FlexNet Manager Suite 2019 R1 or earlier, you can use the Asset Flexera destination to populate endpoint information in FNMS. When you create a Flexera destination, you enable the Flexera-specific content and a set of custom reports and views are created that include the results of these sensors. To send the results of these reports to Flexera, a set of connections in Tanium Connect are automatically created that connect to the SQL database. Flexera communicates with this SQL database to populate information.

Before you begin

- You must have access to Connect with **Connect User** role.
- You must have Connect 4.3.0 or later. With Connect 4.8.0 and later, you can configure your Flexera connections to use views, which is better for large environments.
- To create scheduled actions for the file evidence content, you must have Tanium Administrator privileges.
- You must have an SQL server to use for staging the Flexera data. This database can use Windows or mixed mode authentication. For more information, see Microsoft Docs: Authentication in SQL Server.

Configure Flexera staging database

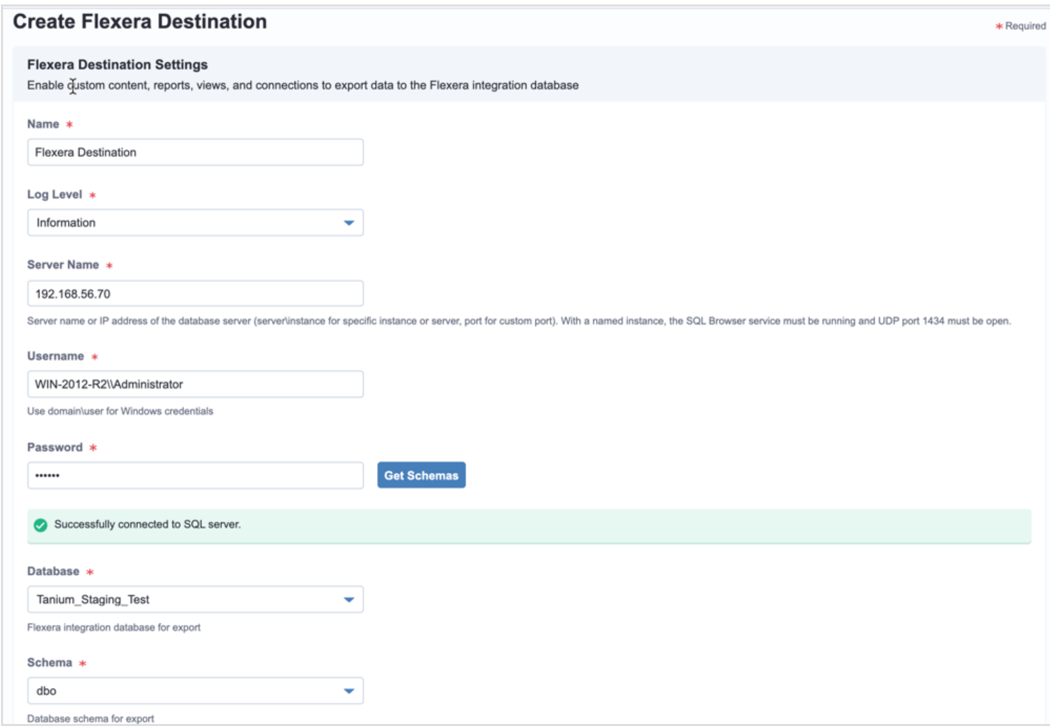
You must have an SQL database configured that implements the required Flexera database schema.

Use the following query to create the required tables and schema for the Flexera staging database: `<Module Server>\services\asset-service\content\flexera\CreateTaniumStagingDatabase.sql`

For more information about configuring Microsoft SQL database to stage data for Flexera, see Tanium Support Knowledge Base: Configuring Microsoft SQL database to stage data for export to Flexera (sign-in required).

Add Flexera destination

1. From the Asset menu, click **Inventory Management > Destinations**.
2. Click **Create Destination > Flexera Destination**.
3. Edit the Flexera settings, including URL and credentials for the SQL server, log level, and the schedule at which you want the export to occur.

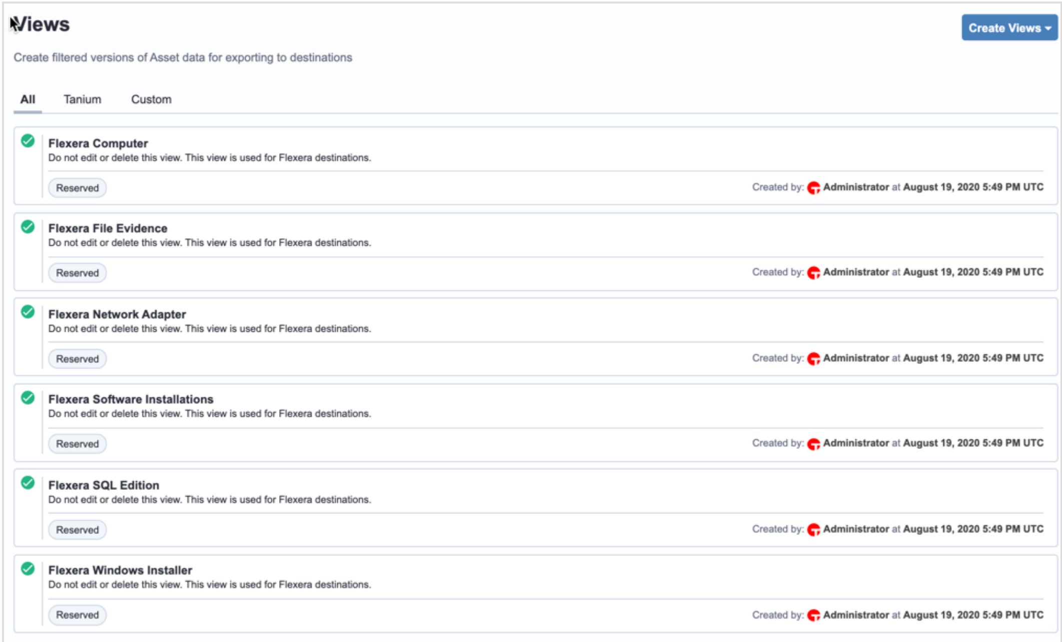
   

   Click **Get Schemas**. When you click this button, a connection is established with the SQL server that looks for databases that match the basic required schema to export Asset data. If a database matches these requirements, it is displayed in the **Database** and **Schema** fields.
4. Click **Create**.

When you add a Flexera destination, the following actions occur:

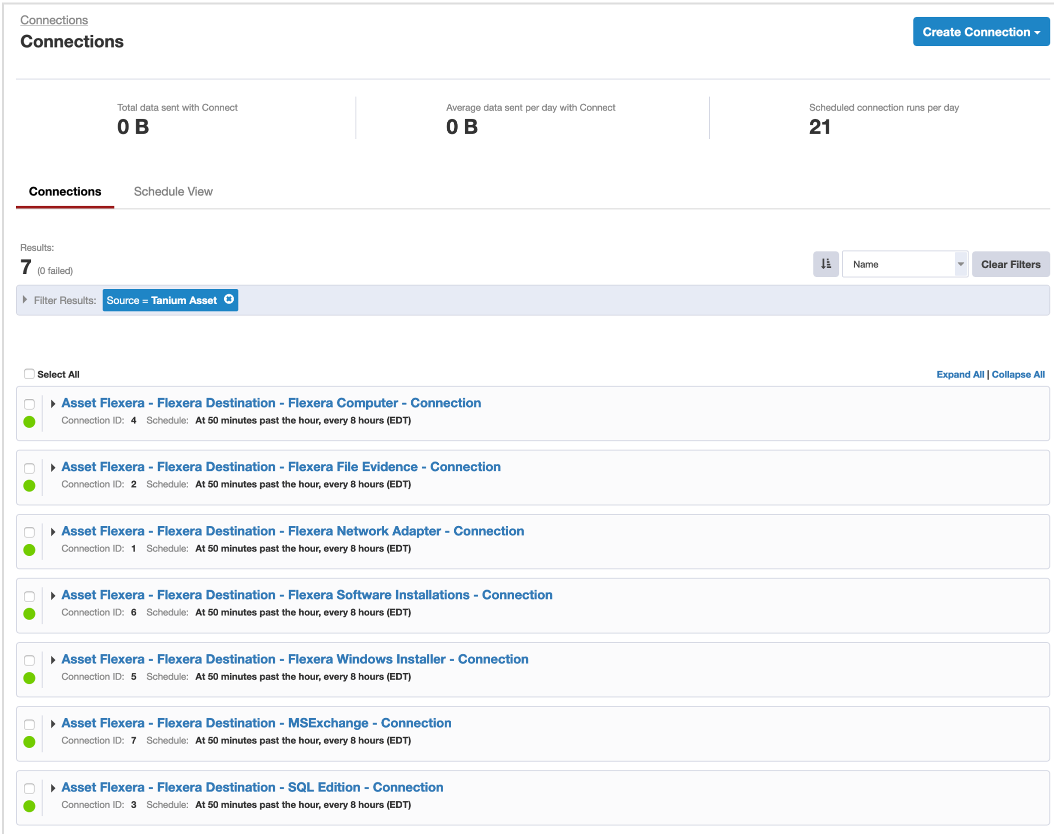
- Additional attributes are added to Asset. These attributes will be pending until the next Tanium import. See View schedule and run import on page 54 for more information.
- Flexera reports are created in Asset. View these reports in the **Reports** section under **Tanium Reports**. Do not delete or modify these reports. Modifying these reports disrupts the Flexera export.

- Flexera views are created.



Do not edit these views because they can be overwritten. If necessary, you can look at the fields that are included in the Flexera views to create a copy of the view that includes different settings.

- For each report, a connection is created in Connect that sends the report data to the SQL server using the views as a source.



Use Connect for all troubleshooting of the data transfer to the SQL server. Each Flexera connection contains information about the schedule and success or failure of the data transfer.

Configure Flexera to receive data from Tanium Asset

Check the contents of your custom reports in Asset and the data that is being exported to the configured SQL server. After the data you want is being exported, configure FlexNet Manager Suite to get data from the database. Work with your Flexera administrator to configure this integration.

(Optional) Enable file evidence content

Asset can integrate with Tanium Index to provide file evidence information to Flexera.

1. Configure Index security exclusions. See Tanium Incident Response User Guide: Before you begin.
2. Install Tanium Index and verify that endpoint file systems are being indexed.

   The **Distribute Tanium Index Tools** , **Distribute Tanium Index Config** and **Start Indexing** packages must be deployed to the endpoints and the **Index Status** sensor should return `Running`. For more information, see Tanium Incident Response User Guide: Install Index and Tanium Incident Response User Guide: Deploy Index tools.
3. When everything is configured, the **Flexera Report File Evidence** custom report begins to get populated with data.

ServiceNow: Tanium Asset integration

You can configure Tanium as a discovery source in ServiceNow, and then create a destination in Asset to export data to ServiceNow with a defined schedule.

Before you begin

- You must be using ServiceNow Jakarta or later release. ServiceNow Software Asset Management Pro is also supported.
- You must have access to both a test and production instance of your ServiceNow Enterprise CMDB.
- You must have a service account for ServiceNow that has elevated privileges. After the initial integration is complete, you can Configure least privilege access in ServiceNow on page 75.

IMPORTANT Test the data export against a copy of your ServiceNow instance before you configure Tanium Asset to export all data to your production instance of ServiceNow. Because the built in identification rules in ServiceNow assume unique computer names or serial numbers, you might need to add one or more identification rules to achieve consistent and expected results.

Prepare ServiceNow to receive Tanium data

1. In ServiceNow, add an entry for Tanium as a choice in the **discovery_source** column of the **cmdb_ci** table. Use `Tanium` as the value in both the **Label** and **Value** fields.
2. (Optional) Work with Tanium Support and your ServiceNow administrator to add tables or update identification rules in ServiceNow. For more information, Contact Tanium Support on page 85.

   Adding identification rules is required if any endpoints in your environment have duplicate serial numbers or computer names. For more information about configuring identification rules, see ServiceNow Documentation: Create or edit a CI identification rule.
3. If you are using ServiceNow Paris 10 or Quebec or later with the Tanium Asset integration, ServiceNow creates duplicate entries for some CIs (such as Network Adapters, Storage Devices, and File Systems) in the ServiceNow CMDB when an export runs. To resolve this, see Fix duplicate entries in ServiceNow with Tanium Asset integration on page 81.

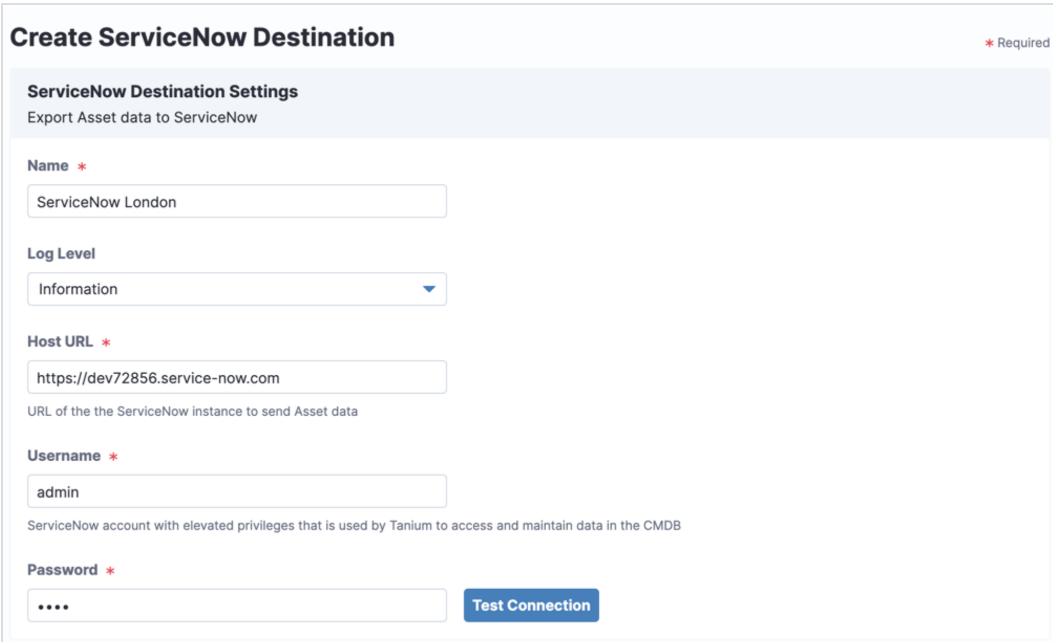
Add ServiceNow as a destination

To enable data to be exported to the ServiceNow CMDB from Asset, enter your ServiceNow Host URL and credentials.

1. From the Asset menu, click **Inventory Management > Destinations**.

2. Click **Create Destination > ServiceNow Destination**.

3. Edit the settings, including the ServiceNow Host URL and credentials, log level, view, and the schedule at which you want the export to occur.



The log level affects the logging in the `job/`*`date_time_job#`*`_servicenow_`*`config#`*`.log` files. If you enable `Trace` level logging on your ServiceNow configuration, numbered subdirectories, for example `job/65`, are created that contain all of the POST and GET requests for that job.
For more information about Cron, see Reference: Cron syntax on page 86.

Exclude computers from exported ServiceNow data

(Optional) When you create a ServiceNow destination, a reserved view is created. The ServiceNow (reserved) view includes all computers. Create a view with filters enabled if you want to narrow the scope of the export.

1. From the Asset menu, click **Views**. Hover over the **ServiceNow (reserved)** view, and click Create Copy.

2. Edit and rename the copy of the reserved view. Add a filter to limit the computers that are exported. In this new view, do not select **Should be filtered out of the view** on any of the filters.

3. From the Asset menu, click **Inventory Management > Destinations > *ServiceNow_Destination***. Click Edit.

4. In the **View** section, select the new view that you created.

5. Click **Update** to save the changes.

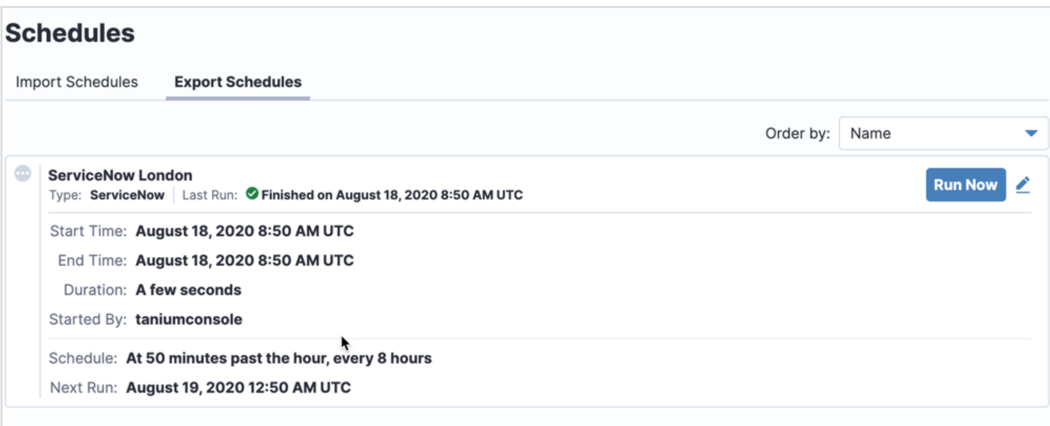
Edit ServiceNow export mappings

(Optional) After you create the ServiceNow destination, you can edit the Asset to ServiceNow mappings.

1. To add an attribute, find the details for the attribute.
    a. From the Asset menu, click **Inventory Management > Attributes**.
    b. Click **Customize Columns**, and select **Table Name** and **Field Name**.
    c. Use the **Filter Items** search to find the attribute.
2. Click **Inventory Management > Destinations > ServiceNow**. Click Edit . In the **ServiceNow Export Mapping** section, add the new mapping. You can also edit individual mappings.
   For more information about the mappings, see Reference: ServiceNow mappings on page 87. Contact Tanium Support to properly edit the ServiceNow export mappings. For more information, Contact Tanium Support on page 85.

Run export

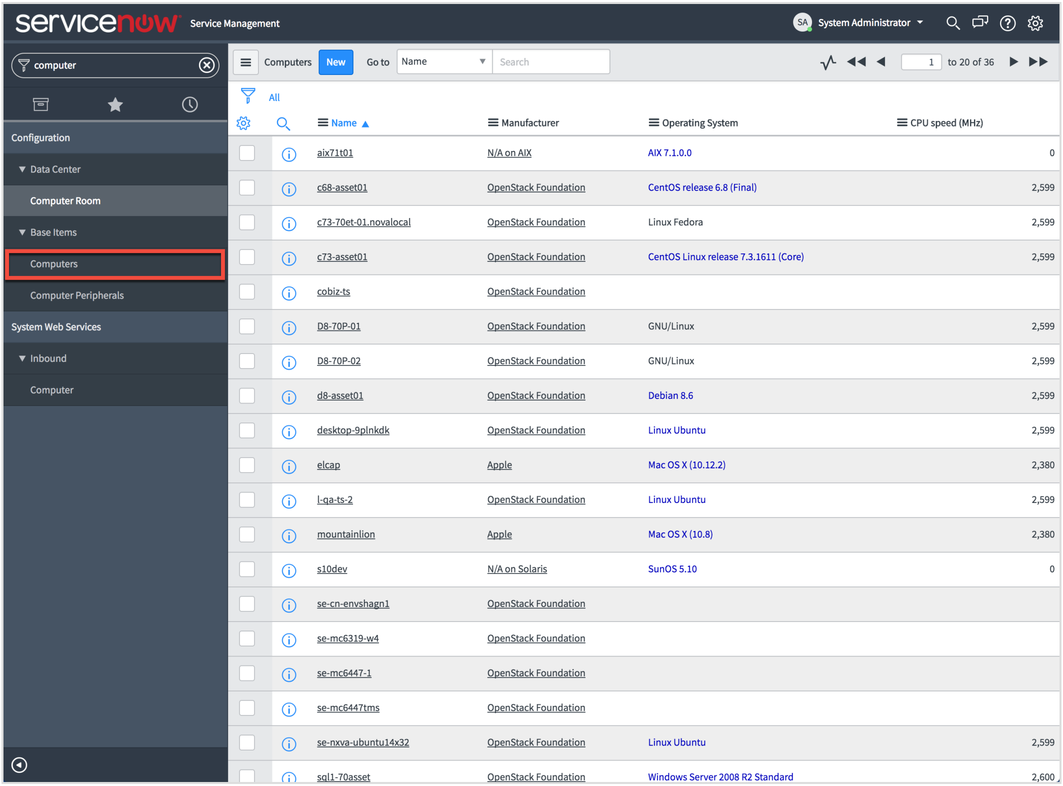
You can run an export to ServiceNow CMDB outside of the configured schedule. From the Asset menu, click **Inventory Management > Schedules> Export Schedules**. Under your ServiceNow destination in the **Asset Export Destinations** section, click **Run Now**.



Check data in ServiceNow

After the Status in the schedule says complete, you can check for the data in ServiceNow CMDB.

1. Sign in to your ServiceNow Enterprise CMDB.
2. Search for an Asset attribute, such as `Computer`.



3. Check the data that got imported into the table.

Configure least privilege access in ServiceNow

After the initial integration with ServiceNow is complete, you can create a role for the Tanium Service Account. This role has an ACL that provides only the necessary permissions.

For more information, see [Tanium Customer Community: Configuring least privilege access for ServiceNow integration sign-in required)](#).

ServiceNow: Service Graph Connector

If you have ServiceNow Orlando Patch 7 or later or Paris Patch 1 or later, you can configure the Service Graph Connector for Tanium Asset app in ServiceNow to import data from Tanium into the ServiceNow CMDB. This integration uses a MID server to collect data from the Tanium Asset API, then sends the data to the ServiceNow data source.

ServiceNow requirements

- ServiceNow Orlando Patch 7 or later or Paris Patch 1 or later.
- ServiceNow MID Server configured. For more information, see [ServiceNow Product Documentation: Installing the MID Server](#).

Tanium requirements

Have the following information available when you configure the Tanium Asset app in ServiceNow:

- A user with the Asset Report Reader role to use with the Tanium Asset API.
- A view defined in Asset to represent the data that you want to send to ServiceNow.
- The URL or IP address of the Tanium Server.

> IMPORTANT If you are using the Tanium Asset app and have an existing ServiceNow destination in Asset, disable the destination after you complete these steps.

Configure secure communication with MID Server

The MID Server is a ServiceNow application that facilitates communication and movement of data between the ServiceNow and external data sources. For the MID Server to communicate with Tanium, you must configure security settings and certificates.

These settings ensure that a valid certificate is used during communication, establishing secure communication between the MID Server and Tanium. If you do not configure these settings, proper certificate validation might not be enforced, resulting in additional security risks that are associated with trusting invalid certificates.

1. Update the `glide.properties` settings on the MID server to enable SSL certificate verification. For more information about updating these settings, see [ServiceNow Product Documentation: General security settings properties](). Set the following values:

   ```
   com.glide.communications.trustmanager_trust_all=false
   com.glide.communications.httpclient.verify_hostname=true
   ```

   After you update these properties, restart the MID Server.

2. Add the Tanium server certificate to the MID Server.

   a. Download the Tanium server certificate. You can download the certificate from the Tanium Server when you log into Tanium from a web browser. For example, in Google Chrome, click the lock icon next to the URL, then click **Certificate**. Drag the image of the certificate to a location on your computer, which makes a local copy of the CER file.

   b. Add the certificate to the MID server. See [ServiceNow Product Documentation: Add SSL certificates for the MID Server]().

Configure Service Graph Connector for Tanium Asset app in ServiceNow

In ServiceNow, configure the Service Graph Connector for Tanium Asset app to connect to the Tanium Asset API. Follow the steps in the **Getting Started** wizard on the **Setup** page.

1. Connect to the Tanium Asset API.
    a. For the **Configure Credentials** step: Specify the Tanium user with Asset Report Reader role. Click **Update**.
    b. Complete the **Configure Connection Details** step:
        - Specify the Tanium Server IP address or host name.
        - Select the MID server on which you configured the Tanium server certificate.
        - Click **Test Connection**.
        - After the connection completes, in **Asset View**, select the copy of the **ServiceNow (reserved)** view.
        - Click **Update**.
2. Schedule imports.
   Configure how often to pull data from the Tanium Asset API. Select the **Active** check box to update the **Repeat Interval** and **Starting** time. By default, concurrent imports are enabled, with a partition size of 15000 records. Click **Update** to save the settings. Click **Execute Now** to run the data import before the next scheduled time.

Monitor and verify data

- Monitor the data imports on the **CMDB Integrations Dashboard** in ServiceNow. Click **CMDB Applications > Tanium Asset Application**. You can view actively running imports, errors, number of records imported, and so on.
- After an import completes, you can check for the data in ServiceNow CMDB. Search for an Asset attribute, such as `Computer` and verify that Tanium data was added.

Troubleshooting Asset

To collect and send information to Tanium for troubleshooting, collect log and other relevant information.

Collect logs

The information is saved as a compressed ZIP file that you can download with your browser.

1. From the Asset **Overview** page, click Help .
2. Click **Troubleshooting > Collect**. When the ZIP file is ready, you can download the `tanium-asset-support-[timestamp].zip` file to your local download directory.
3. Contact Tanium Support to determine the best option to send the ZIP file. For more information, see Contact Tanium Support on page 85.

Update service account log level

1. From the Asset **Overview** page, click Settings , then the **Advanced** tab.
2. Change the **Service Account Log Level**.
3. Click **Save**.

Configure report timeout period

Configure the report timeout period to increase or decrease the amount of time that Asset tries to load a report page before presenting a timeout error. To troubleshoot the error, see Troubleshoot reports on page 78.

1. From the Asset **Overview** page, click Settings , then the **Advanced** tab.
2. Specify the **Report Timeout**. After this timeout period, Asset presents an error.
3. Click **Save**.

Troubleshoot reports

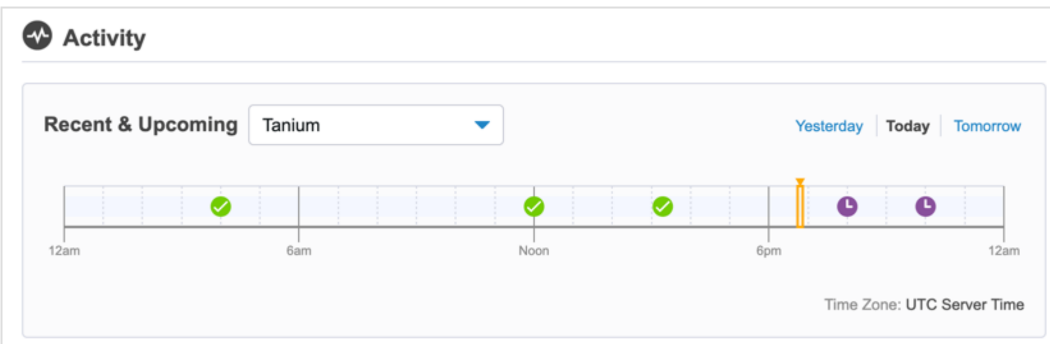
If you see a report timeout error message when trying to view a report, consider the following resolutions or workarounds.

- Try to view the report again.
- Configure report timeout period on page 78.
- Look at the data using a view.
  - Create a view that includes the assets. For more information, see Configuring views on page 57.
  - Export the data to Connect or another destination. For more information, see Exporting data to destinations on page 64.

Troubleshoot asset data exports and imports

View status of imports and exports

On the Asset **Overview** page in the Activity section, you can view a timeline of the recent imports and exports.



The timeline contains each import, with one of the following statuses:

- : Scheduled
- : Successful
- : Error

View import and export logs

In the ZIP file that you download from the **Help** section, you can view logs for the data imports and exports. These logs are in the `job` directory:

ServiceNow exports

ServiceNow export logs are named with the following format: `job/`*`date_time_job#_servicenow_config#`*`.log`. If you enable `Trace` level logging on your ServiceNow configuration, numbered subdirectories, for example `job/65`, are created that contain all of the POST and GET requests for that job. See Add ServiceNow as a destination on page 72 for more information about configuring logging.

Asset data imports

You can use these logs to view details about the scheduled runs that are occurring to import asset data from Tanium into your Asset database. Tanium data import logs are named with the following format: `job/date_time_job#_tanium_1.log`. For data you are importing from a database, import logs are named with the following format: `job/date_time_job#_database_1.log`. To change the log level for imports, see Configuring sources on page 45.

Disable imports and exports

Data imports and exports run on a configured schedule by default. You can disable this schedule to resolve issues, or run the import or export manually.

1. From the Asset menu, go to **Inventory Management > Schedules**.
2. Hover over the import or export schedule that you want to disable and click Edit.
3. Clear the **Run this connection on a defined schedule** setting. Click **Update**.
4. If you want to manually run an import or export, go to the **Inventory Management > Schedules** page. Hover over the import or export schedule and click **Run Now**.

Fix `Error: MULTIPLE_DUPLICATE_RECORDS` error message in ServiceNow log file

If you see an `Error: MULTIPLE_DUPLICATE_RECORDS` error message in the ServiceNow log file, contact Tanium Support to verify that the ServiceNow export mapping configuration is configured correctly. For more information, Contact Tanium Support on page 85.

1. From the Asset menu, click **Inventory Management > Destinations**.
2. Click Edit next to the destination.
3. In the **ServiceNow Export Mapping** section, add the following property to the **cmdb_ci_computer** mapping:

```
identificationFields: [
  [
    "name",
    "serial_number"
  ]
],
```

4. Click **Update**.

Fix missing information in export to ServiceNow with Tanium Asset integration

If Asset has finished exporting data to ServiceNow using the Tanium Asset integration and some endpoints are missing or are missing information, work with Tanium Support to review trace logs and make necessary changes.

1. Update the ServiceNow destination log level to trace.
    a. From the Asset menu, click **Inventory Management > Destinations**.
    b. Click Edit next to the destination.
    c. Set the **Log Level** to **Trace**.
    d. Click **Update**.
2. Run the export again.
    a. From the Asset menu, click **Schedules > Export Schedules**.
    b. Click **Run Now** next to the schedule.
3. Collect logs on page 78.
   The logs include a `ServiceNow > Jobs` folder that contains job IDs for the requests and responses made to ServiceNow.
4. Contact Tanium Support on page 85 for assistance reviewing logs and making necessary changes.
   Search the logs for the following errors.

Error	Explanation
`ABANDONED`	Generally jobs are abandoned because of a large number of errors. These errors might occur because of the difference between identifying an endpoint in Asset and ServiceNow. Asset identifies endpoints based on multiple attributes, and ServiceNow (by default) identities endpoints based on a unique name and serial number. This error might also indicate missing required fields in ServiceNow because of data class customization or a different business rule.
`TIMED`	Timed out jobs are usually caused by a business rule or other database issue within ServiceNow. If the `duration` field in the log file is around 300, Asset might have exported too much data to ServiceNow. You can decrease the job size using an `API Settings` mapping.
`logContextID`	`logContextId` indicates that an entry in the job was logged. Usually errors are categorized as `ABANDONED` or `TIMED`.

Fix duplicate entries in ServiceNow with Tanium Asset integration

If you are using ServiceNow Paris 10 or Quebec or later with the Tanium Asset integration, ServiceNow creates duplicate entries for some CIs (such as Network Adapters, Storage Devices, and File Systems) in the ServiceNow CMDB when an export runs. This is a known ServiceNow issue. Update the ServiceNow system properties to workaround this issue.

1. In ServiceNow, enter `sys_properties.list` in the **Filter navigator**.
2. In **System Properties**, click **New**.
3. Enter `glide.identification_engine.dependent_items_bulk_query` as the **Name** and `true` as the **Value**.
4. Click **Submit**.

Troubleshoot software inventory and usage monitoring

To return inventory and usage data, endpoints must have the correct version of Asset and the **swmgr** tool installed.

1. In Interact, ask the following question:
   `Get Endpoint Configuration - Tools Status Details from all machines`
2. For the **Asset** tool name, confirm that the installed version is correct.
3. For the **swmgr-cx** tool name, confirm that the installed version is 2.0.1.0 or later.

Back up and restore the Asset database

You can back up or restore all configuration and Asset data. You might want to create a backup before making configuration updates. You must have the Tanium Administrator role to perform a backup operation.

1. From the Asset **Overview** page, click Settings, then the **Database** tab.
2. To create a backup of the Asset database, click **Back Up Database**.
3. Use the backups.
   - Download backup on local computer: Click Download Database Backup.
   - Restore database backup: Click Restore Database Backup. Before you restore, verify that no Asset destination exports or Tanium Connect connection runs are running. When you run the restoration, the current Asset database is backed up, the Asset service is shut down, the selected database is restored, and the Asset service is restarted. Any jobs that are running during the restoration process do not complete.
     If you want to restore the database on a different Module server, contact Tanium Support. For more information, see Contact Tanium Support on page 85.
   - Delete a backup: Click Delete Database Backup.

Remove unneeded data from the Asset database

Delete assets

Asset automatically purges stale assets from the database that have not been seen by Asset after the configured stale data age or based on a purge report. For more information, see Configure data retention and vacuuming on page 83.

You can manually remove assets from your asset database earlier than the stale data age. In a report that shows the assets you want to remove, select the rows and click **Delete**.

If the asset you deleted responds to the Asset saved questions in a subsequent load, the asset is re-added to the database.

Delete attributes

To delete an attribute, you must disable the attribute, then remove references.

1. Disable the attribute.
    a. From the Asset menu, click **Inventory Management > Attributes**.
    b. Click Edit next to the attribute that you want to disable.
    c. Clear the **Enabled** setting. Click **Save**.
2. Remove references to the attribute from all reports, views, permissions, and ServiceNow destinations.
    a. Click Delete next to the attribute that you want to remove.
    b. A list of known locations where the attribute is used is displayed. Click the link for each item, and edit the item to remove the use of the attribute.
    c. Click **Refresh** to list the remaining references.
3. After you remove the references, type the name of the attribute to confirm its deletion. Click **Delete Attribute**.

The attribute is deleted from the Asset database on the next load of data from the data source.

Configure data retention and vacuuming

You can configure data retention and automatic vacuuming on the Asset database.

1. From the Asset **Overview** page, click Settings, then the **Advanced** tab.
    - To purge stale assets that have not been seen by Asset from the database, enable **Purge Stale Assets**. Then, indicate the age of stale data to remove. The minimum number is seven days. The default is 180 days. The maximum number is unlimited days.

      TIP

      If you set the maximum number to a large number, consider filtering old assets from reports and views.
    - To purge assets based on a report, enable **Purge by Report** and select the report. To select a report, you must be an Asset Administrator and not assigned to a user group. The report must meet the following criteria:
        - Is not a summary or reserved report.
        - Only has attributes from the Asset column.
        - Has a filter with attributes that do not include parentheses. For example, the report can include **Asset ID** but not **Device ID (Physical Disk)**.
    - To adjust the trigger and amount of work done during automatic vacuuming, adjust the **Cost Limit** and **Scale Factor** values. The Postgres VACUUM operation reclaims storage that is occupied by dead tuples. By default, the database is vacuumed when 1% of tuples are considered dead, and the cost limit (amount of work per vacuum cycle) is set to 1000.
2. Click **Save**.

Monitor and troubleshoot key vendors with tracked products

The following table lists contributing factors into why the Key Vendors with Tracked Products metric might be lower than expected and corrective actions you can make.

Contributing factor	Corrective action
Products are not tracked	In **Software Inventory & Usage > All Products**, track products from key vendors.
Lack of knowledge on which products to track	Use the hidden **Asset SIU** sensors for ad hoc validation of endpoint usage. Based on that informal validation, identify products to track.

Monitor and troubleshoot unused tracked software

The following table lists contributing factors into why the Unused Tracked Software metric might be lower than expected and corrective actions you can make.

Contributing factor	Corrective action
Extraneous software is being tracked	• Only track software that is valuable to the business • Only track software when tracking is in accordance with your company's software asset management charter
No formal software reclamation guidelines exists	Work with your asset management team to clearly define when software should be reclaimed. This could be based on last used date or overall usage over time. An example might be consistent low usage of an expensive piece of software.

Uninstall Asset

1. From the Main menu, click **Tanium Solutions**. Under Asset, click **Uninstall**. Click **Proceed with Uninstall** to complete the process.
2. Remove Asset Tools from your endpoints. To see which endpoints have the file evidence tools installed, ask the question: `Get Asset File Evidence Status from all machines`. If you want to clean the artifacts from your endpoints, Contact Tanium Support on page 85.
3. A backup `asset-files` folder gets created as part of the uninstall process. You can keep or delete this folder. If any other Asset artifacts remain on your Module Server, Contact Tanium Support on page 85..
4. Remove Asset saved questions. You can remove saved questions that meet all the following conditions:
   - Owned by the service account you configured for Asset
   - AND the name of the saved question starts with `Asset`
   - AND is in the `Asset` content set

5. Remove Asset scheduled actions. Delete the `Asset Deploy Collect Active Directory Info` scheduled action that is created by the service account and running the `Asset Collect Active Directory Info` package.

6. Remove Asset action group. After the action group is empty, you can delete the `Tanium Asset` action group.

Contact Tanium Support

To contact Tanium Support for help, sign in to https://support.tanium.com.

Reference: Cron syntax

A quick reference to Cron syntax follows. You can use Crontab to build a Cron expression.

```
┌───────────── minute
│ ┌─────────── hour
│ │ ┌───────── day of month
│ │ │ ┌─────── month
│ │ │ │ ┌───── day of week
│ │ │ │ │
│ │ │ │ │
* * * * *
```

Each asterisk is a field that must be included in the Cron expression. The field value can either be an asterisk (any value) or one of the following values:

Valid values for Cron fields

Field	Value
minute	0-59
hour	0-23
day of month	1-31
month	1-12
day of week (Sunday is 0 and 7)	0-7

Reference: ServiceNow mappings

The following table describes how Asset attributes are mapped to ServiceNow attributes. The attributes are available on all Tanium Client supported operating systems, unless otherwise indicated.

Category	ServiceNow Attribute	Asset Attribute	Comments
Computer	Computer Name	Computer Name	Filter-normalized
	Serial Number	Asset Computer Serial Number	
	Operating System	Asset Operating System	
	OS Version	Asset OS Version	
	OS Service Pack	Asset Service Pack	Not available on macOS, Linux, and Solaris
	Manufacturer	Asset Manufacturer	Not available on Solaris and AIX. Lookup to core_company
	Model	Asset Model	Lookup to cmdb_model
	OS Domain	Asset Domain Name	Filter-normalized
	DNS Domain	Asset Domain Name	
	RAM	Asset RAM	Normalized
	Virtual	Asset Is Virtual	
	IP Address	Tanium Client IP Address	
	CPU Name	Asset CPU	
	CPU Manufacturer	Asset CPU Manufacturer	Not available on AIX
	CPU Speed	Asset CPU	Normalized
	CPU Count	Asset CPU	
	CPU Core Count	Asset CPU	
	CPU Type	Asset CPU	

Category	ServiceNow Attribute	Asset Attribute	Comments
Storage Device	Name	Asset Physical Disk	
	Device ID	Asset Physical Disk	
	Serial Number	Asset Physical Disk	
	Manufacturer	Asset Physical Disk	Lookup to core_company
	Model ID	Asset Physical Disk	Lookup to cmdb_model
	Storage Type	Asset Physical Disk	Filter-convert to ServiceNow values
	Device Interface	Asset Physical Disk	Filter-convert to ServiceNow values
	Size	Asset Physical Disk	Normalized
File System	Name	Asset Logical Disk	Not available on Solaris and AIX
	Mount Point	Asset Logical Disk	Not available on Solaris and AIX
	Label	Asset Logical Disk	Not available on Solaris and AIX
	Size	Asset Logical Disk	Not available on Solaris and AIX. Normalized.
	Free Space	Asset Logical Disk	Not available on Solaris and AIX. Normalized.
	Media Type	Asset Logical Disk	Not available on Solaris and AIX. Convert to ServiceNow values.
	File System	Asset Logical Disk	Not available on Solaris and AIX. Convert to ServiceNow values.
Network Adapter	Name	Asset Network Adapter	
	IP Address	Asset Network Adapter	
	Netmask	Asset Network Adapter	
	MAC Address	Asset Network Adapter	
	MAC Manufacturer	Asset Network Adapter	Lookup to core_company
	DHCP Enabled	Asset Network Adapter	

Category	ServiceNow Attribute	Asset Attribute	Comments
Software Package	Name	Asset Installed Applications	
	Version	Asset Installed Applications	
	Manufacturer	Asset Installed Applications	Lookup to core_company
	ServiceNow Software Identifier (Key)	Asset Installed Applications	Filter-convert to ServiceNow values. This is not the license key and is only used by ServiceNow.